# Module 3

# Introduction to Wearables- Time Series

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THE MICHAEL J. FOX FOUNDATION FOR PARKINSON'S RESEARCH
Founding funders:
UKRI Medical Research Council
EDMOND J. SAFRA PHILANTHROPIC FOUNDATION

# What We'll Do Today

- **Lecture:** Short introduction to Parkinson's disease, wearables and time-series data

**Tutorial 2 / Challenge** — Intro to **time-series data**: clean signals and extract features from **ECG** and **accelerometer** data.

- Start with pandas: tabular health records (EHR).
- Work with real biomedical signals (ECG + accelerometer).
- Preprocessing: missing data, smoothing, normalisation, filtering, downsampling.
- ECG: R-peak detection, RR intervals, heart rate.
- Accelerometer: windowing, time-domain features, vector magnitude.
- End-of-day task: link wearable features to health records.



Cecilia Rodriguez

# Lecture Plan

## 1 Parkinson's disease

- What Parkinson's is
- No disease-modifying treatment yet
- The PPMI cohort
- Clinical measures: UPDRS and MoCA
- Biological markers: DaTscan and CSF α-synuclein

## 2 Why wearables

- Why digital markers
- How Parkinson's is monitored today
- StrivePD and PKG
- Predicting Parkinson's up to 7 years early

## 3 What they measure

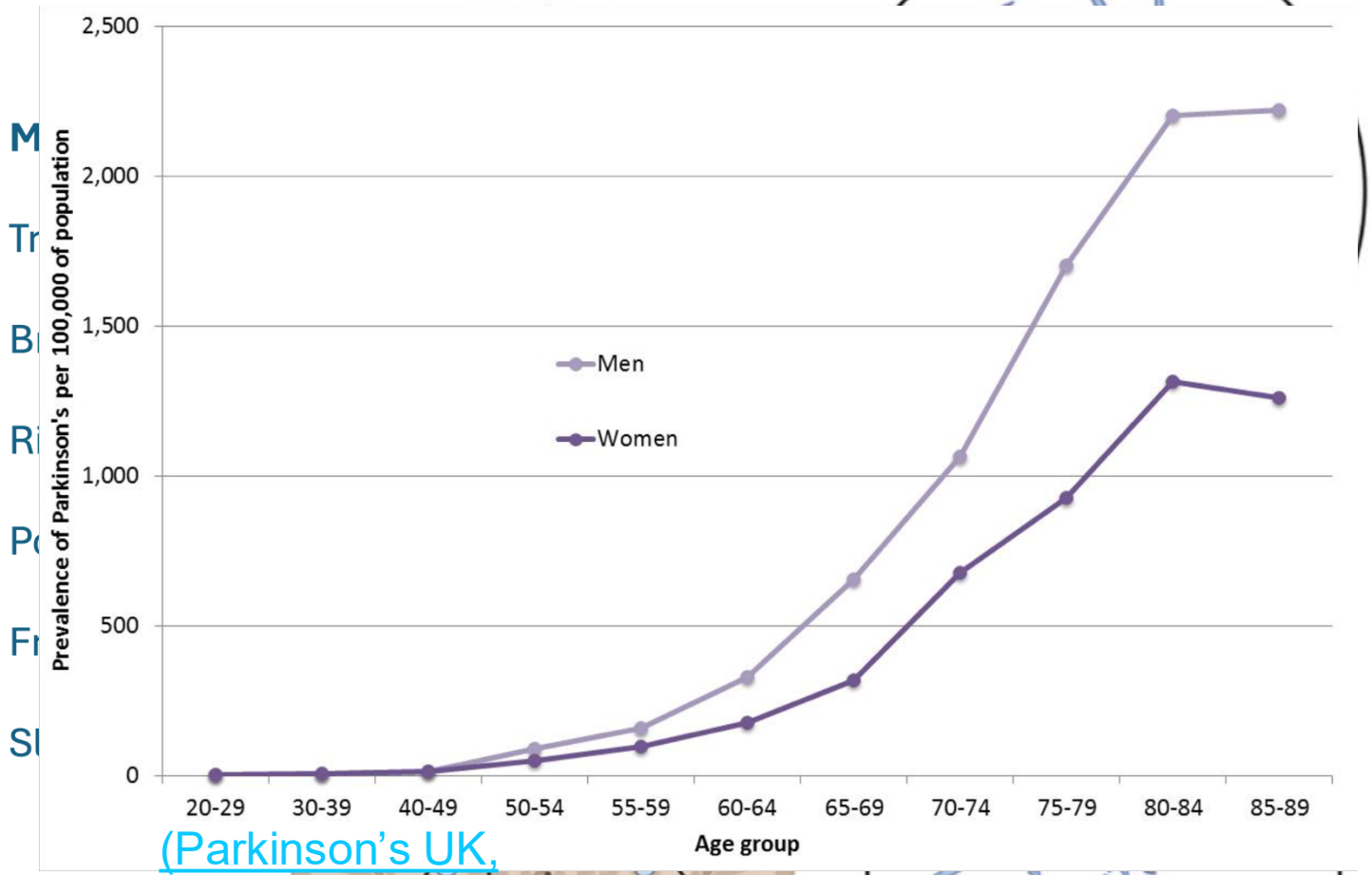
- IMU: accelerometer and gyroscope
- Tremor and bradykinesia
- PPG
- ECG
- Time series in neuroscience

## 4 From signal to numbers

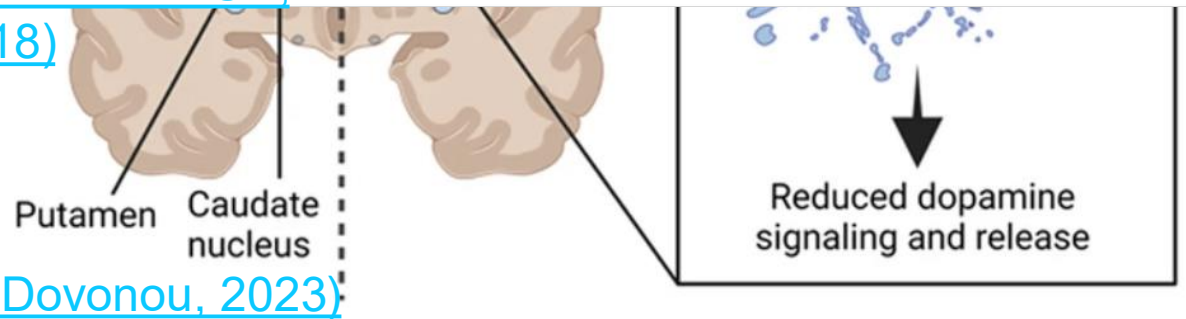
- Missing data
- Frequency domain
- Moving average
- Band-pass filtering
- Windows and features

# Quick intro to Parkinson's

- A **progressive neurological disorder**
- Caused by the **degeneration** of dopamine-producing neurons in the substantia nigra
- Characterized by hallmark **motor symptoms**
- **1 in every 37 (2.7%)** people diagnosed at some point in life
- **No cure**, only symptomatic treatment




(Parkinson's UK, 2018)




(Dovonou, 2023)

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# No established disease-modifying treatments for Parkinson's disease

THE LANCET
Neurology

REVIEW · Volume 20, Issue 7, P559-572, July 2021

## Progress towards therapies for disease modification in Parkinson's disease

Nirosen Vijiaratnam, FRACP [a,b] · Prof Tanya Simuni, MD [c] · Prof Oliver Bandmann, PhD [d] · Prof Huw R Morris, PhD [a,b] · Prof Thomas Foltynie, PhD [a,b]

### Summary

The development of interventions to slow or halt the progression of Parkinson's disease remains a priority for patients and researchers alike. To date, no agents have been shown to have unequivocal evidence of disease-modifying effects in Parkinson's disease. The absence of disease-modifying treatments might relate not only to inadequate approaches for the selection of therapeutic candidates but also to insufficient attention to detail in clinical trial design. Better understanding of Parkinson's disease pathogenesis associated with advances in laboratory models, the use of objective biomarkers of disease progression and target engagement, and a focus on agents known to be safe for human use, alongside the use of precision medicine approaches, should together greatly increase the likelihood for successful identification of disease-modifying treatments for Parkinson's disease.

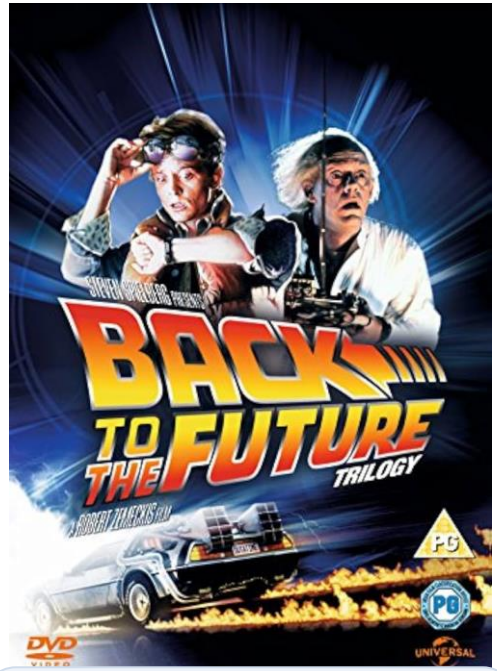
- Parkinson's is primarily managed with medication that **controls symptoms**
- Dopaminergic medication alleviates symptoms, but its effects diminish over time & cause side effects

**No** established treatments to **stop/slow/modify** the disease course!

# Parkinson's Progression Markers Initiative

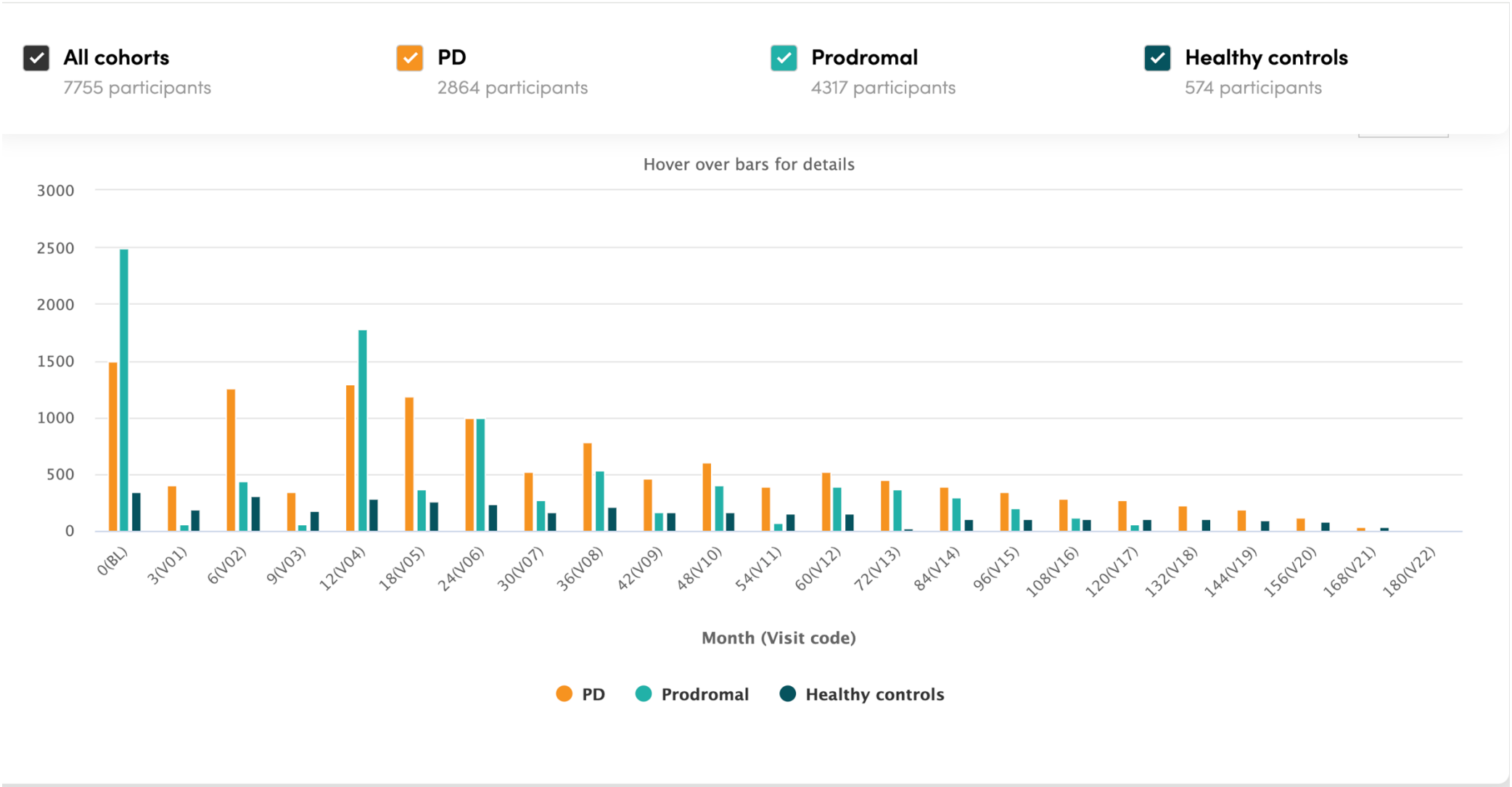
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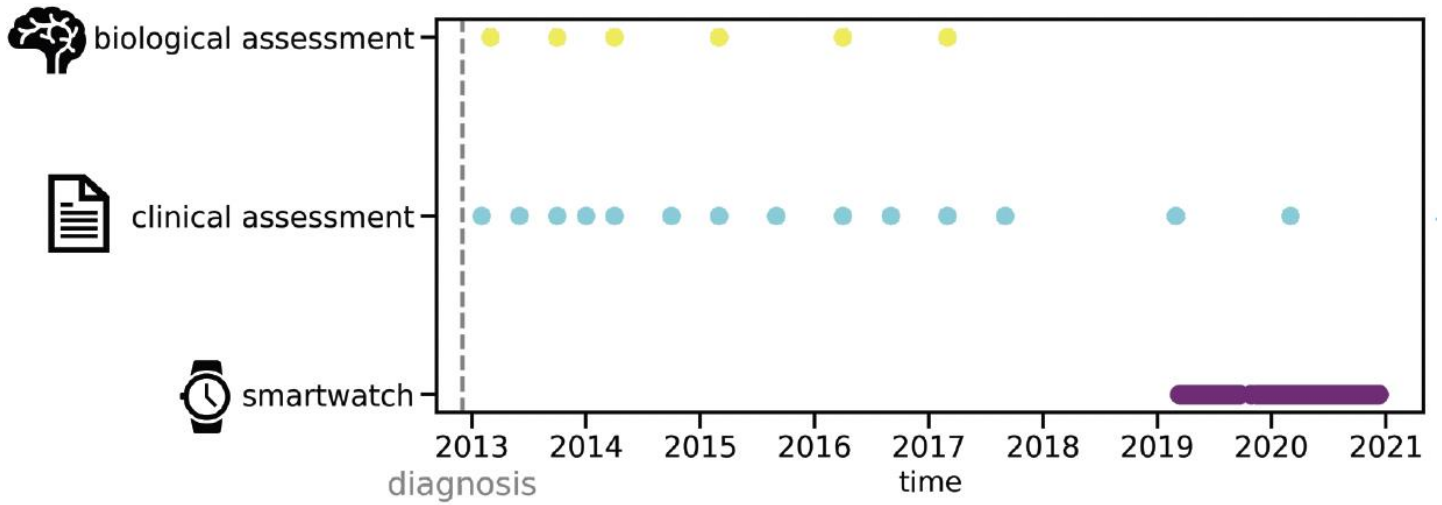




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# Parkinson’s Progression Markers Initiative



# Data Modalities




**Clinical Assessments:**
Unified Parkinson's Disease Rating Scale, Montreal Cognitive Assessment, Drug Prescriptions (Workshop 3)

**Biological Assessments:**
Neuroimaging (Dopamine Transporter Scan), Cerebrospinal Fluid (Seed Amplification Assay), Blood

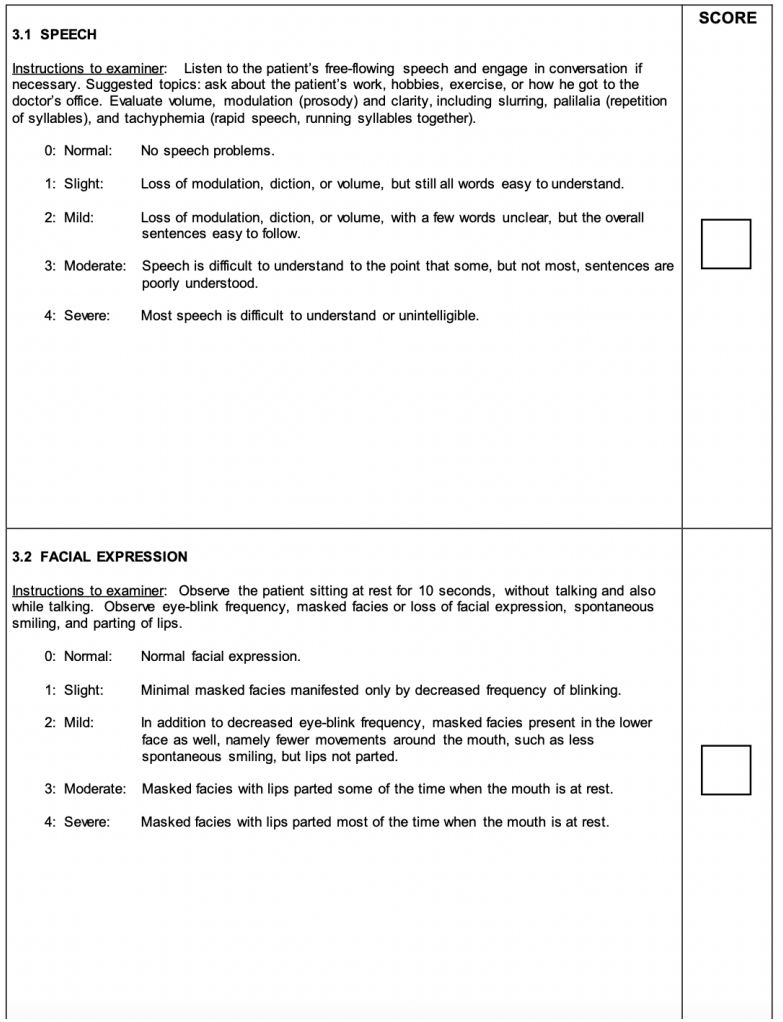
**Digital Assessments:**
Smartwatch (Workshop 2)

# Unified Parkinson's Disease Rating Scale (UPDRS)

Part I: Non-Motor Aspects of Experiences of Daily Living – Questionnaire
Part II: Motor Aspects of Experiences of Daily Living - Questionnaire
**Part III: Motor Examination – Clinician scored**
Part IV: Motor Complications (dyskinesias and motor fluctuations) – Questionnaire

Each item has 0 - 4 ratings:
0: Normal
1: Slight
2: Mild
3: Moderate
4: Severe

18 items
Score range: 0–132
32 and below is mild
59 and above is severe



| | SCORE |
|---|---|
| **3.1 SPEECH**<br>Instructions to examiner: Listen to the patient's free-flowing speech and engage in conversation if necessary. Suggested topics: ask about the patient's work, hobbies, exercise, or how he got to the doctor's office. Evaluate volume, modulation (prosody) and clarity, including slurring, palilalia (repetition of syllables), and tachyphemia (rapid speech, running syllables together).<br>0: Normal: No speech problems.<br>1: Slight: Loss of modulation, diction, or volume, but still all words easy to understand.<br>2: Mild: Loss of modulation, diction, or volume, with a few words unclear, but the overall sentences easy to follow.<br>3: Moderate: Speech is difficult to understand to the point that some, but not most, sentences are poorly understood.<br>4: Severe: Most speech is difficult to understand or unintelligible. | ☐ |
| **3.2 FACIAL EXPRESSION**<br>Instructions to examiner: Observe the patient sitting at rest for 10 seconds, without talking and also while talking. Observe eye-blink frequency, masked facies or loss of facial expression, spontaneous smiling, and parting of lips.<br>0: Normal: Normal facial expression.<br>1: Slight: Minimal masked facies manifested only by decreased frequency of blinking.<br>2: Mild: In addition to decreased eye-blink frequency, masked facies present in the lower face as well, namely fewer movements around the mouth, such as less spontaneous smiling, but lips not parted.<br>3: Moderate: Masked facies with lips parted some of the time when the mouth is at rest.<br>4: Severe: Masked facies with lips parted most of the time when the mouth is at rest. | ☐ |

# Montreal Cognitive Assessment (MoCa)

## Scoring the MoCA Test

The total score on the M
domain is broken downs

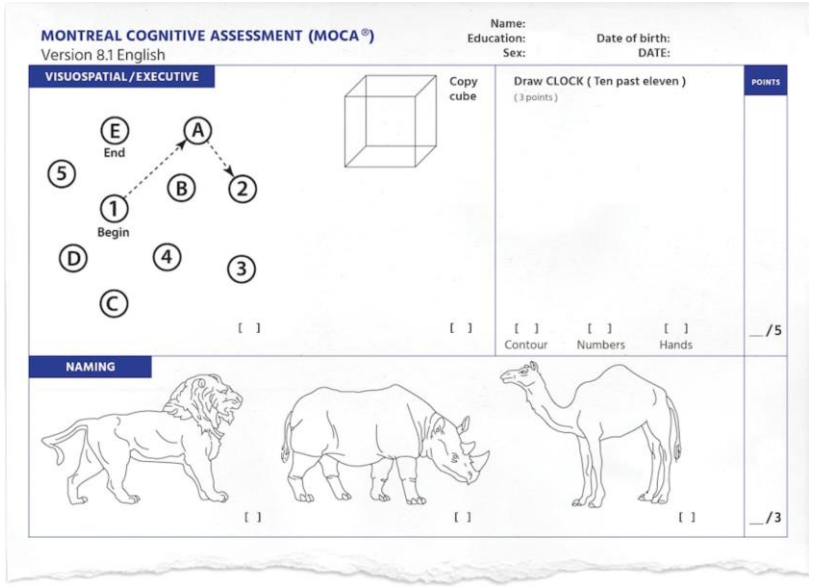
| Domain |
|---|
| Executive/visuospatial function |
| Naming |
| Attention |
| Language |
| Abstraction |
| Recall |
| Orientation |
| TOTAL |

Because a person's educ
tasks, **1 point** is added tc
formal education.

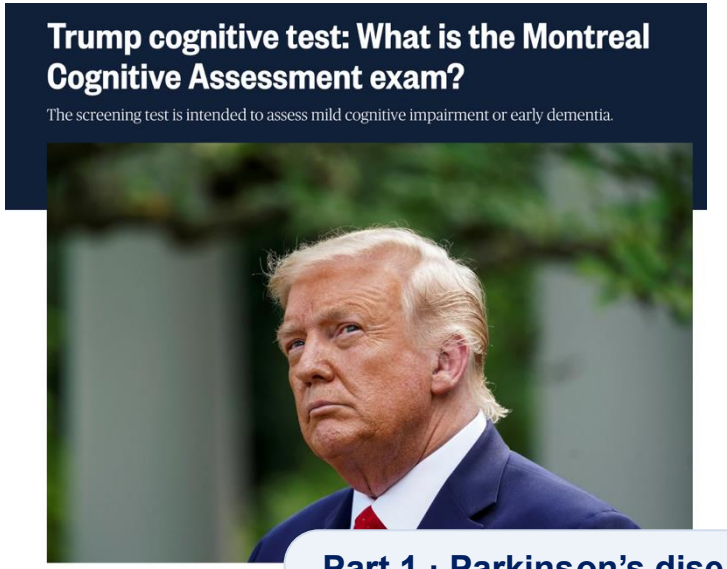
### Interpreting the Results

After tallying the MoCA scores, the results can be interpreted as follows:

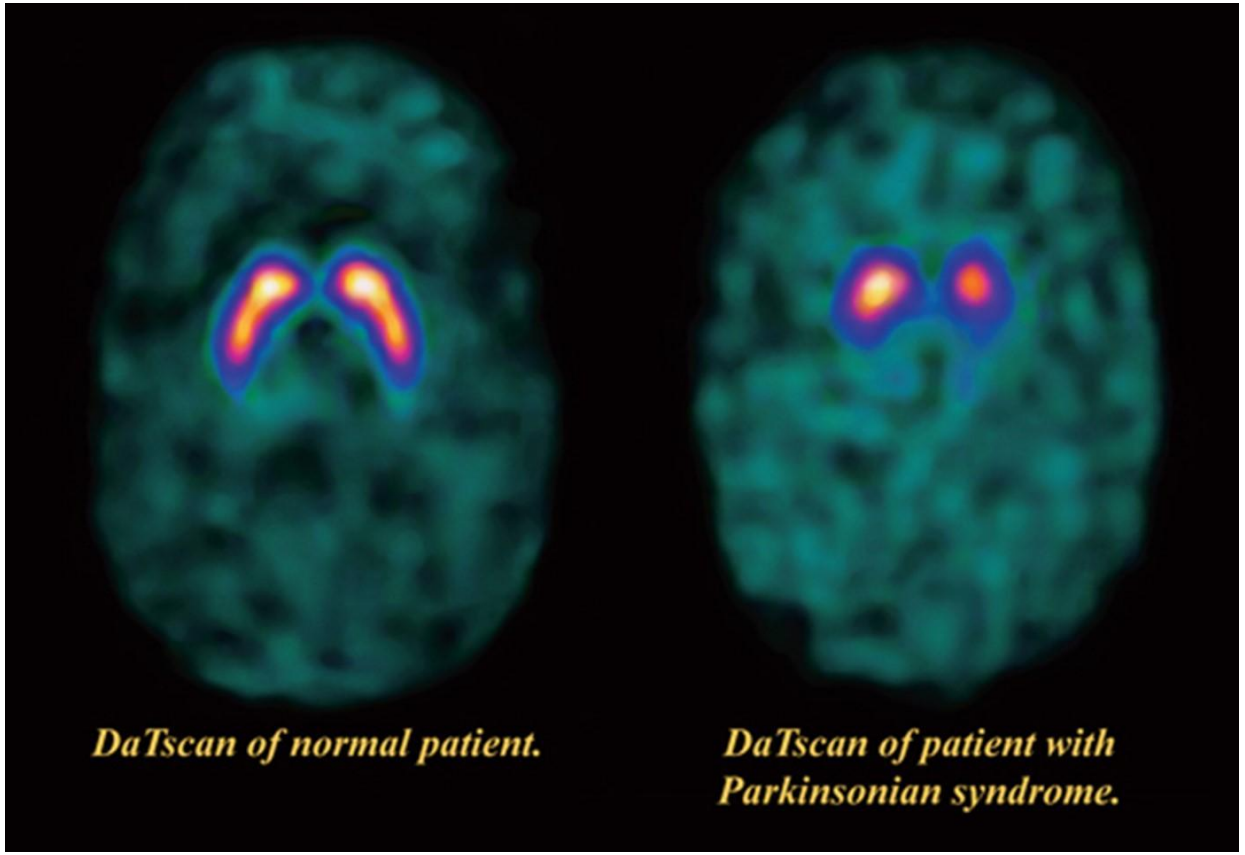
| Interpretation | Score range |
|---|---|
| Normal cognition | 26–30 points |
| Mild cognitive impairment | 18–25 points |
| Moderate cognitive impairment | 10–17 points |
| Severe cognitive impairment | Under 10 points |




30/30



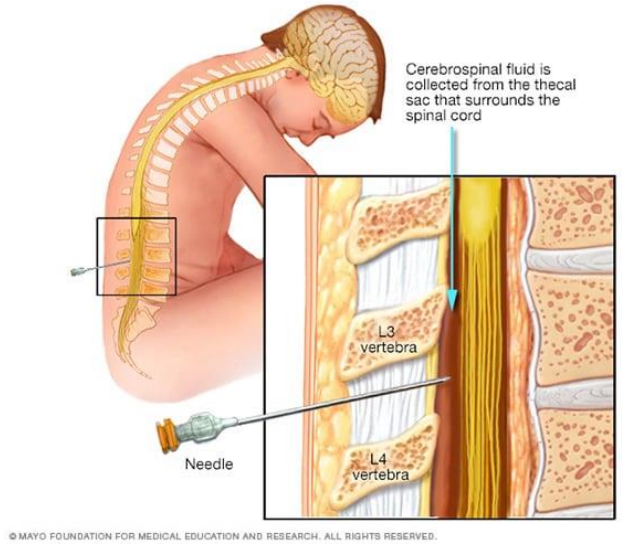
# DaTscan



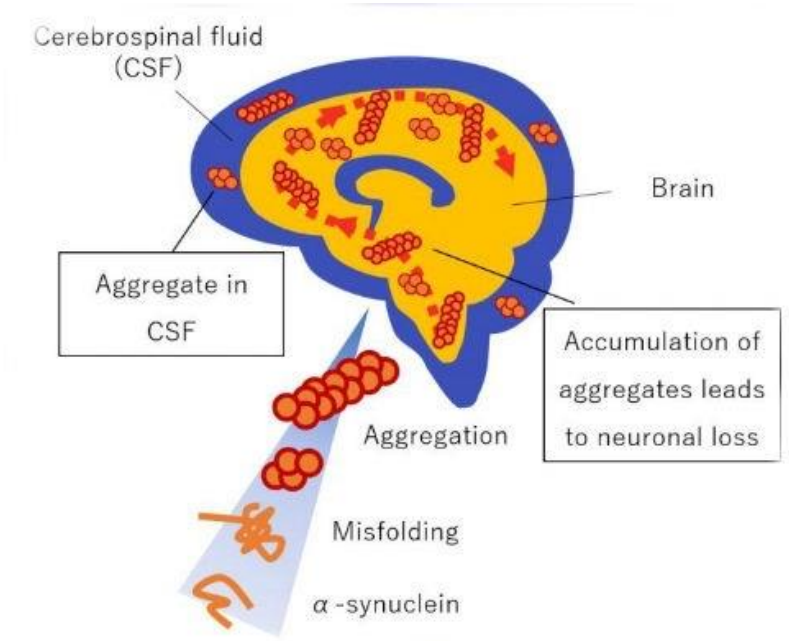

- **DATScan:** visualization of dopamine transporter (DaT) function in the striatum (caudate and putamen) using SPECT brain imaging => can detect **nigrostriatal neurodegeneration**

**PROBLEM: time-consuming, expensive & invasive tests!**

# CSF test: α-Synuclein Seeding Assay Using RT-QuIC

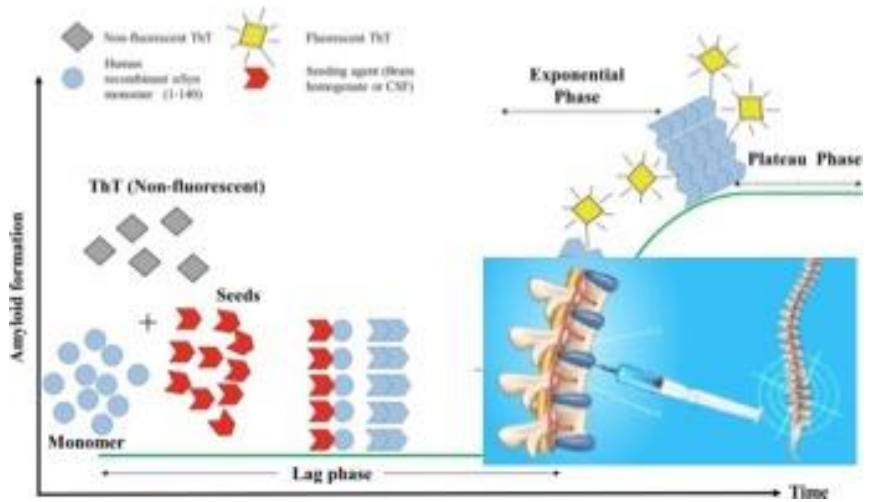


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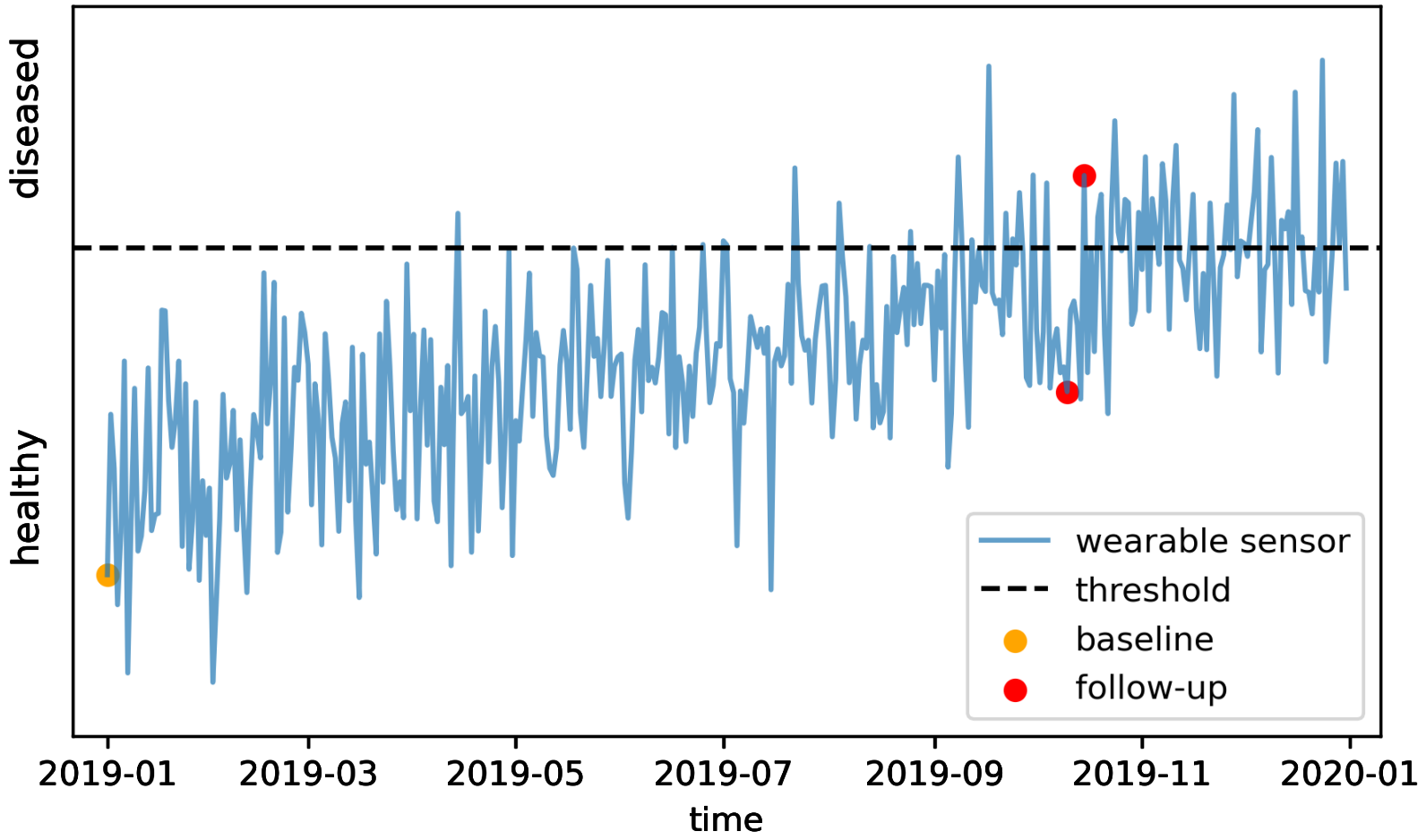

- **Cerebro-spinal fluid assays:** the quantification of α-synuclein (αSyn) in cerebrospinal fluid (CSF) has been proposed as a diagnostic biomarker for Parkinson's disease. **Invasive:** requires a lumbar puncture to collect CSF

**PROBLEM: time-consuming, expensive & invasive tests!**

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# Why should we focus on digital markers?





# Monitoring: What do we currently use?

## Clinical visits: UPDRS

**3.1 SPEECH** — SCORE

Instructions to examiner: Listen to the patient's free-flowing speech and engage in conversation if necessary. Suggested topics: ask about the patient's work, hobbies, exercise, or how he got to the doctor's office. Evaluate volume, modulation (prosody) and clarity, including slurring, palilalia (repetition of syllables), and tachyphemia (rapid speech, running syllables together).

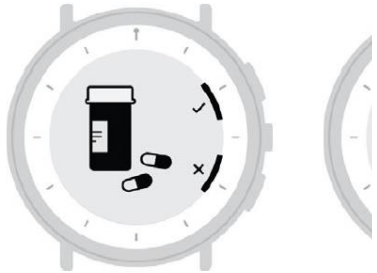
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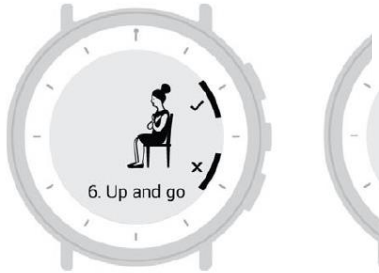
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## Active virtual assessment















## Passive digital motor features

# Rune Labs scores FDA 510(k) to monitor Parkinson's symptoms on Apple Watch

The StrivePD app uses Apple's Movement Disorder API to track tremors and uncontrolled body movement.

By **Emily Olsen** | June 14, 2022 | 11:42 am

SHARE Share 20.6K

# NeuroRPM obtains FDA 510(k) for Parkinson's monitoring on Apple Watch

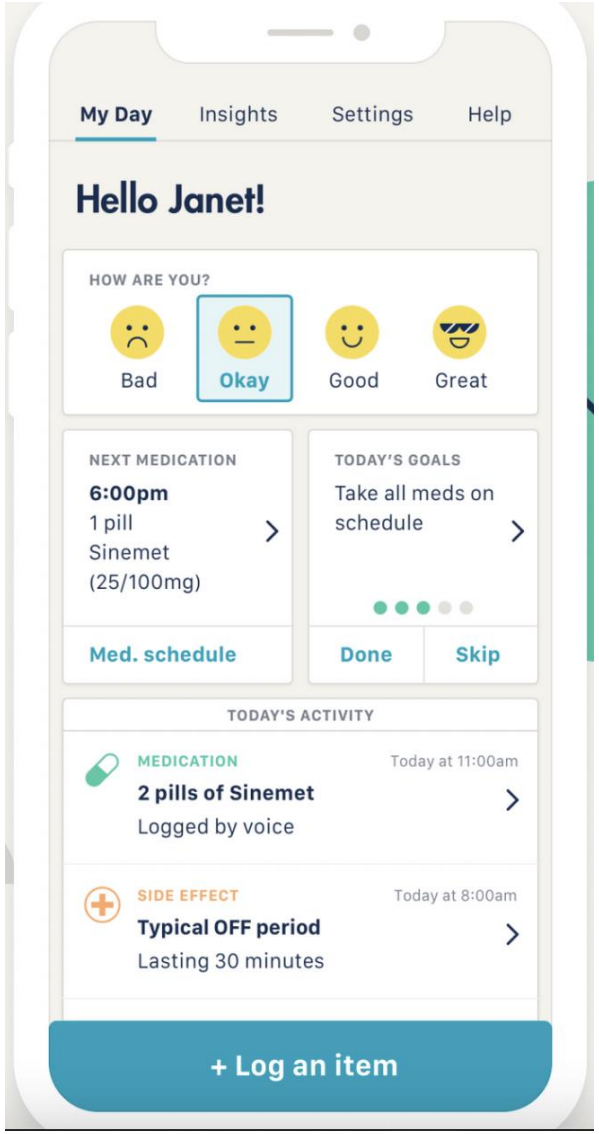
The AI-powered app allows for remote monitoring of common Parkinson's disease symptoms such as bradykinesia, tremor and dyskinesia via an Apple Watch.

By **Jessica Hagen** | March 24, 2023 | 11:18 am

SHARE Share 268

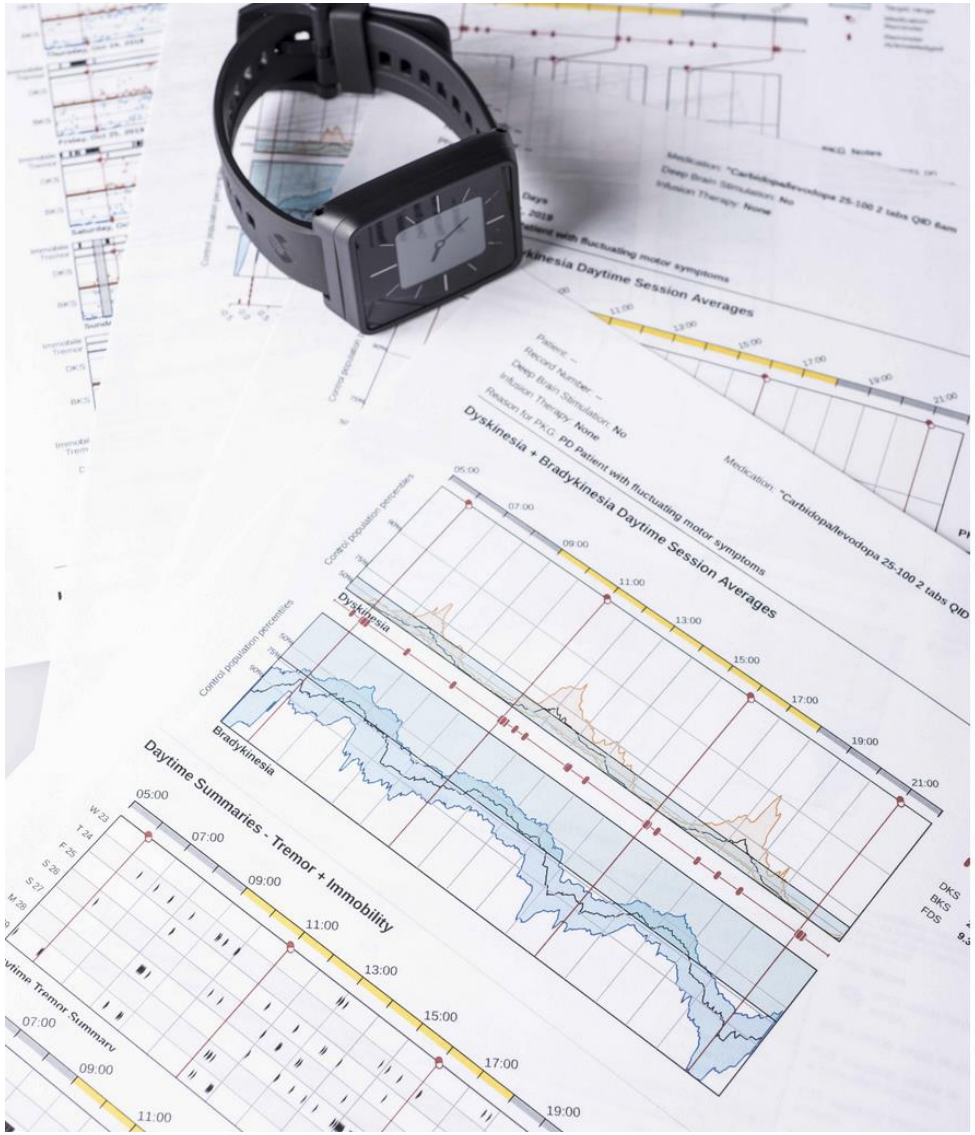
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## StrivePD




https://www.strive.group/

## PKG




https://pkgcare.com/

# Smartwatch data can predict, up to 7 years in advance, who is going to develop Parkinson's disease.

biobank uk



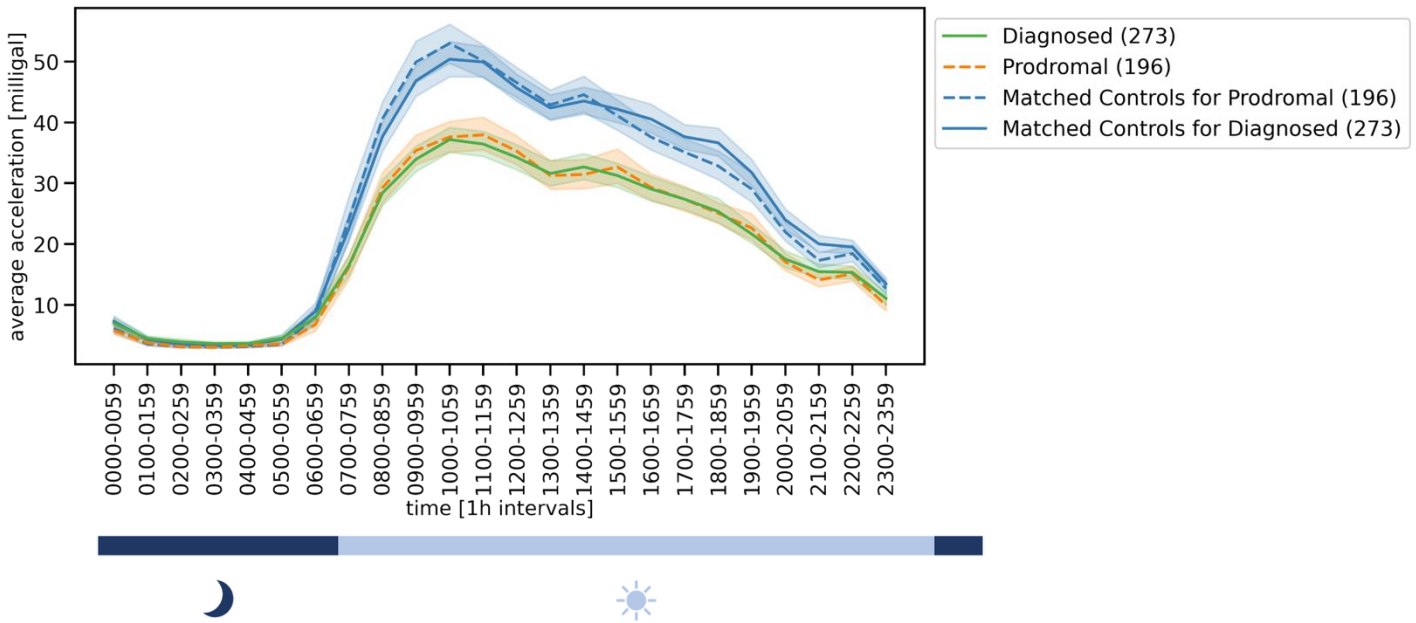
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Article | Published: 03 July 2023

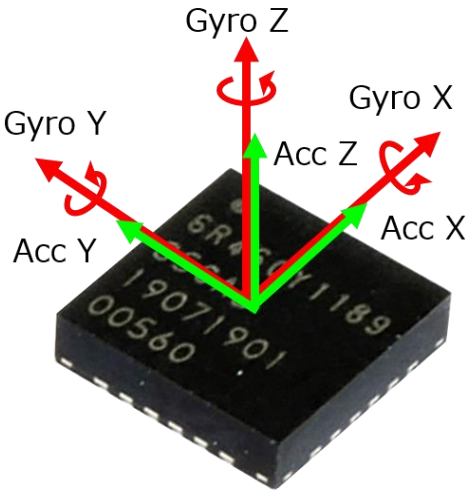
## Wearable movement-tracking data identify Parkinson's disease years before clinical diagnosis



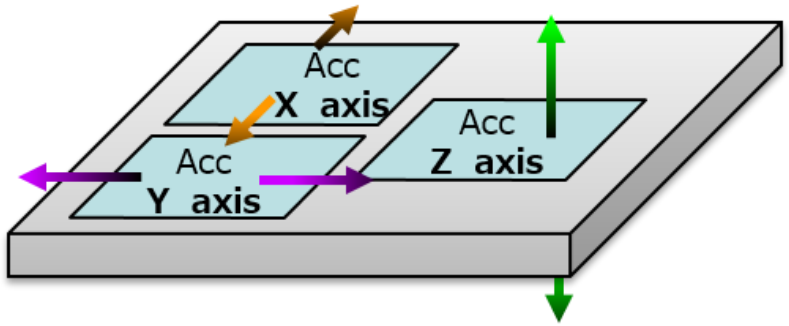

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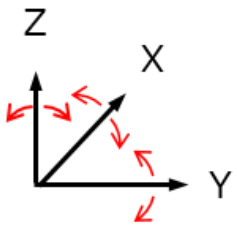
# Wearables Time-Series Data

**6-dimensional inertial measurement unit**, or **IMU**.

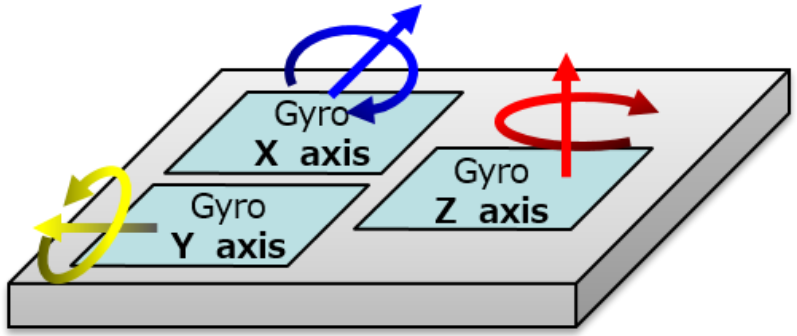


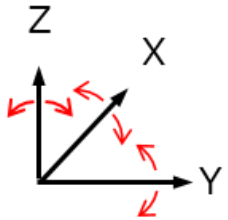

### 3-axis accelerometer



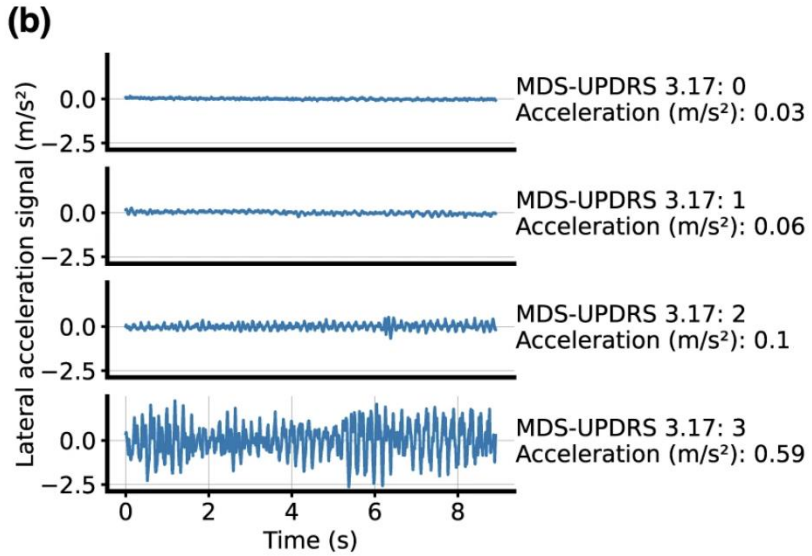

### 3-axis gyroscope

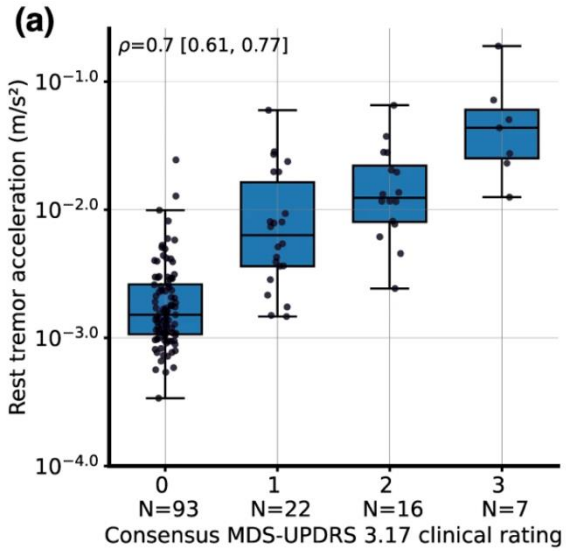




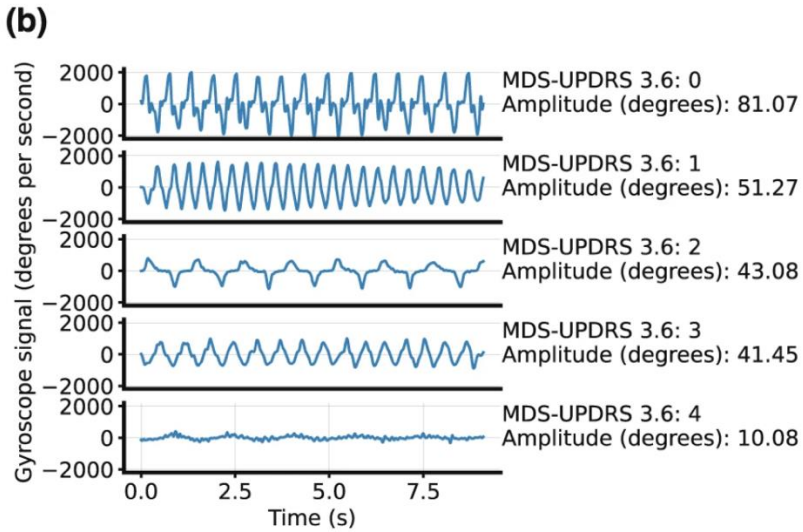

# From Raw Signals to Clinical Meaning: Tremor and Bradykinesia

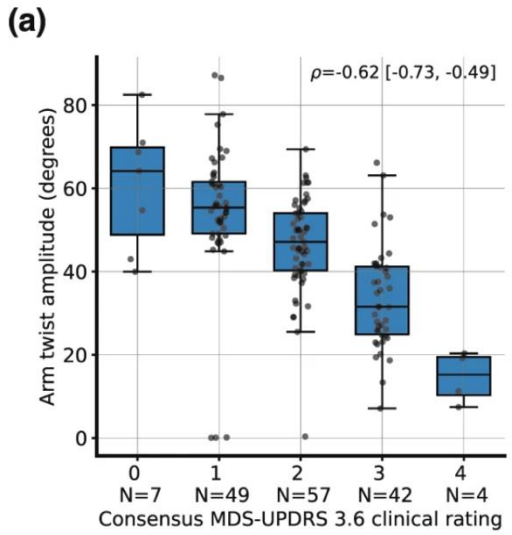


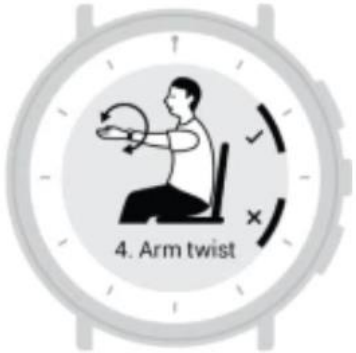









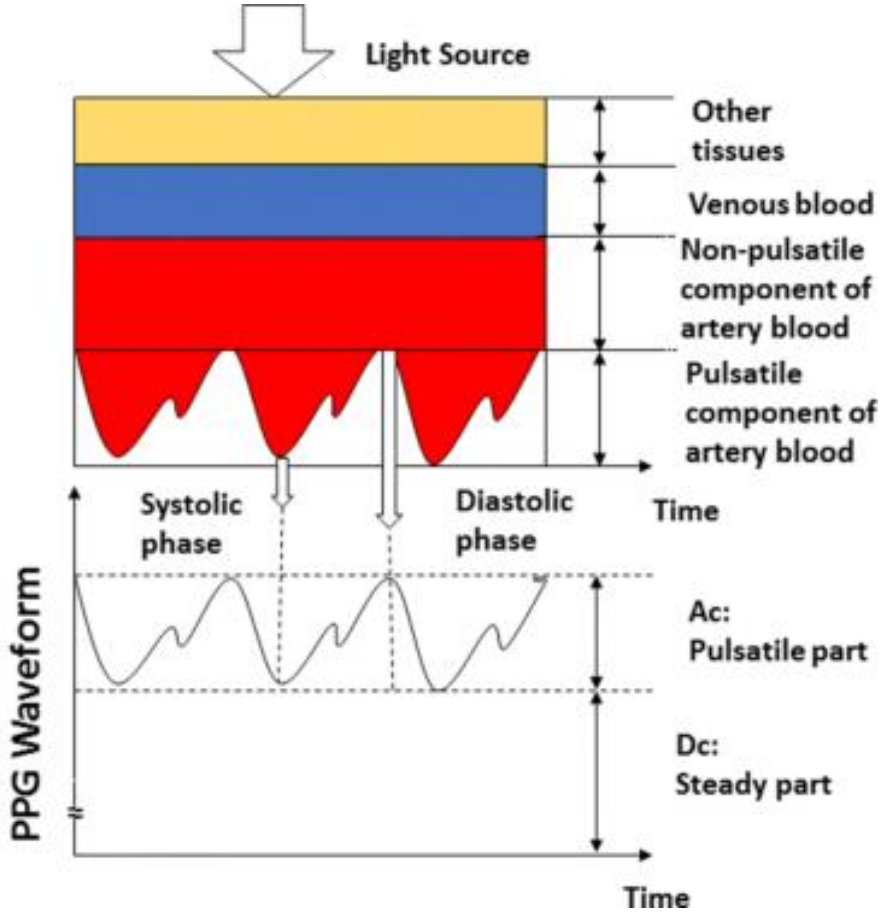
# Beyond IMU - Smartwatch



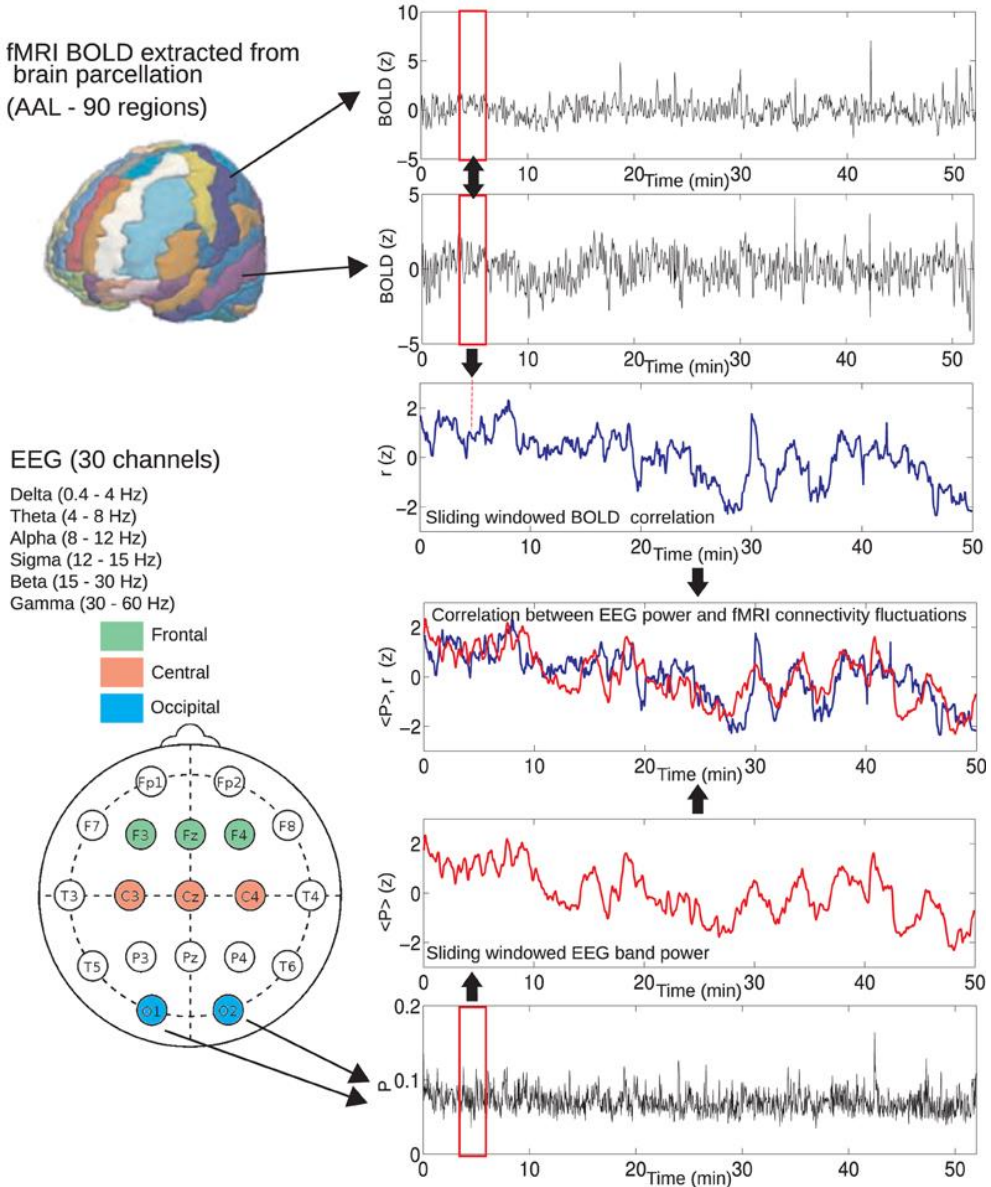
Multi-sensor device:
Accelerometer
Ambient temperature
Gyroscope
Humidity
**Photoplethysmography (PPG)**

verily



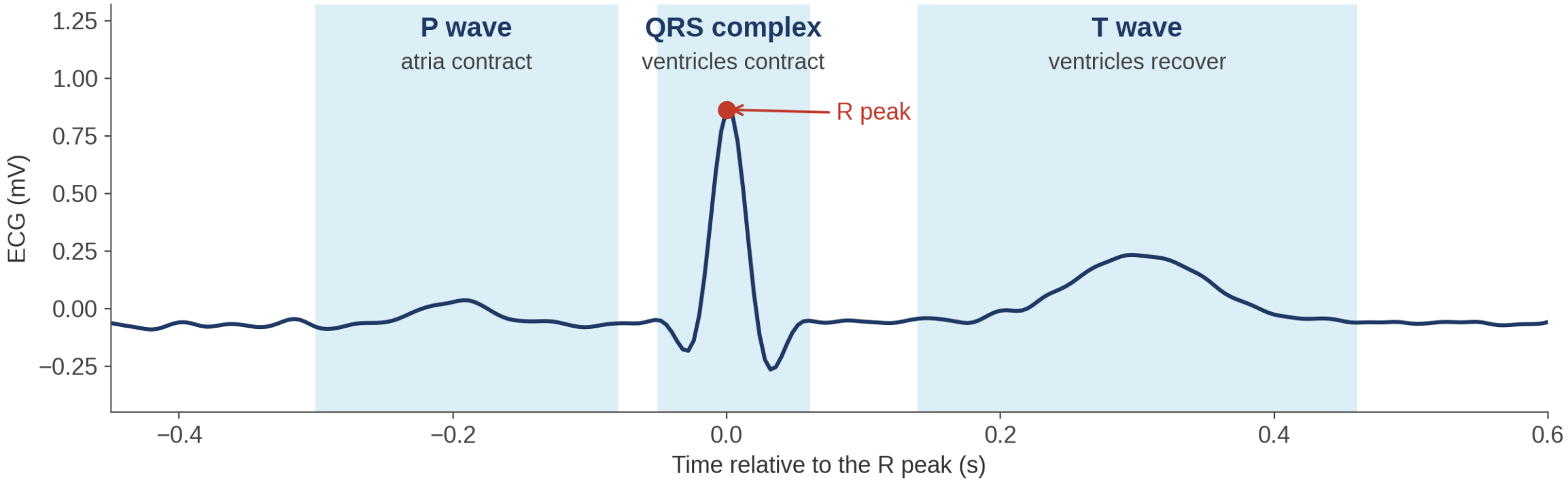



# Beyond Smartwatch:
# Time-Serie Data very common in Neuroscience



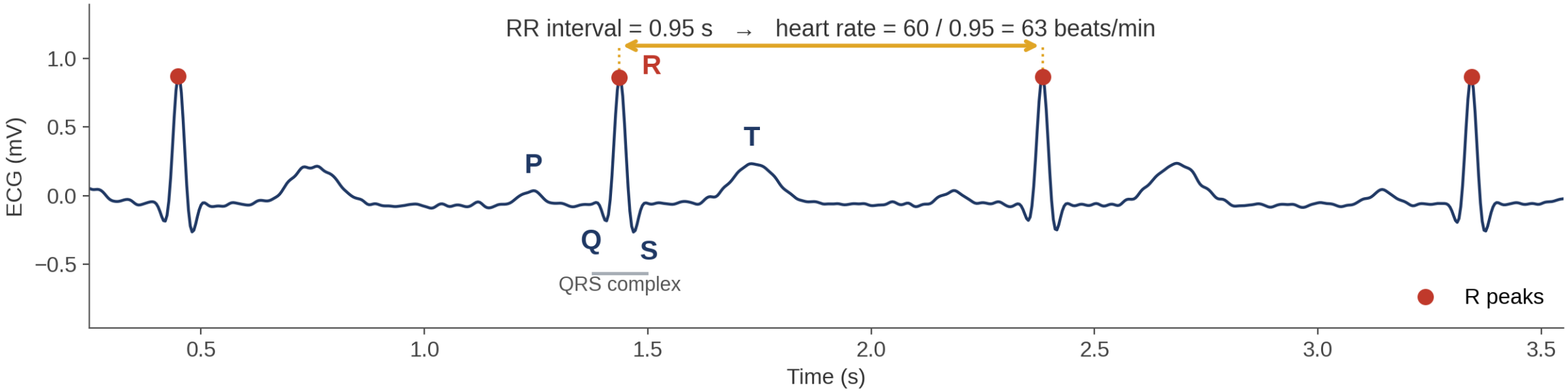

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# ECG: the shape of a heartbeat



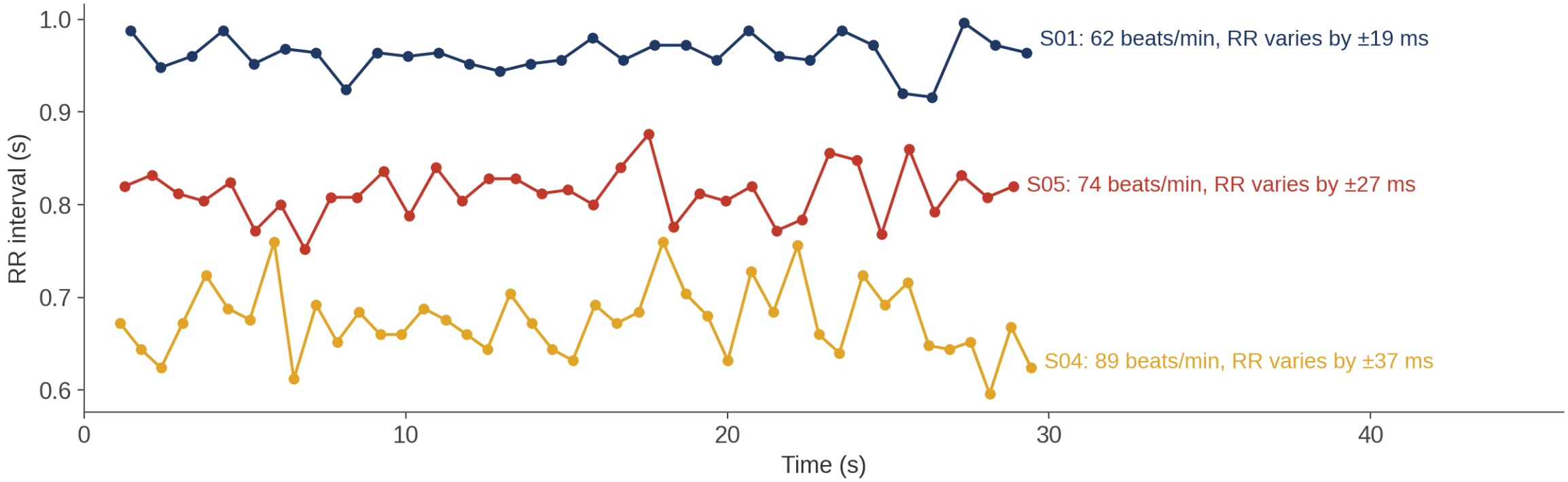

Every heartbeat draws the same three waves. The tall, sharp R peak is the easiest to find automatically.

# ECG: from R peaks to heart rate



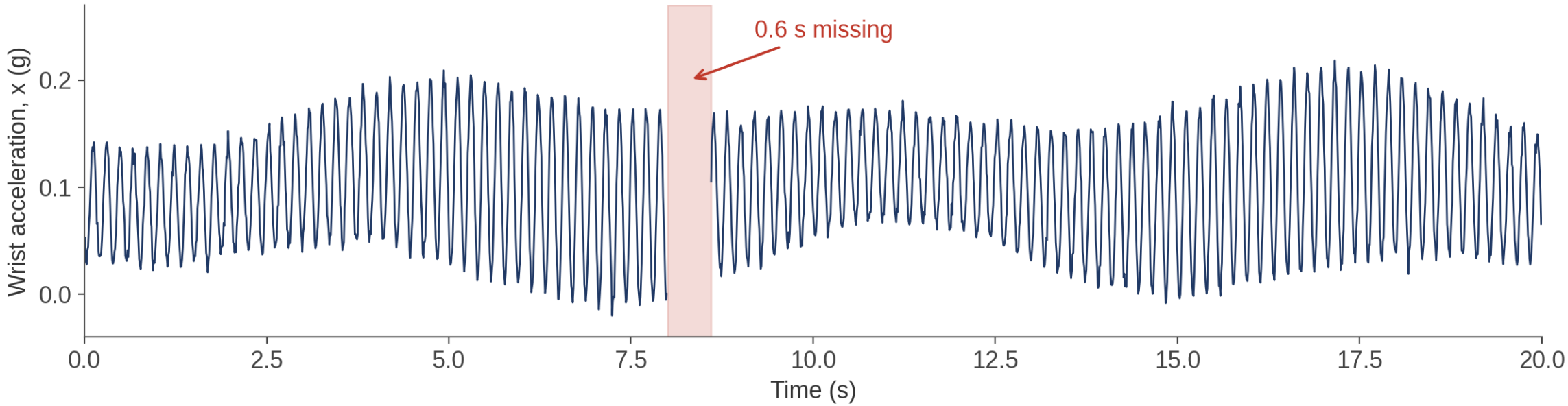

Find the R peaks, measure the time between them, and heart rate follows: 60 / RR.

# Heart rate variability



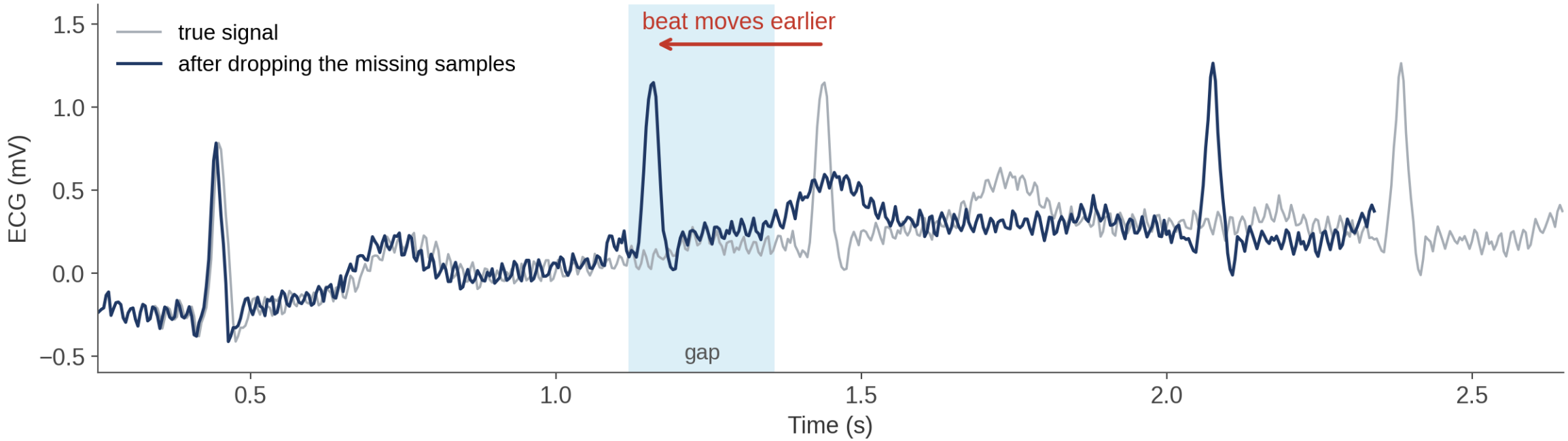

The time between beats is never perfectly constant. This beat-to-beat variation reflects the autonomic nervous system, which Parkinson's can affect.

# Missing data: real recordings have gaps




Electrodes peel off and watches lose contact. One missing value is enough to break an average, so gaps must be handled first.

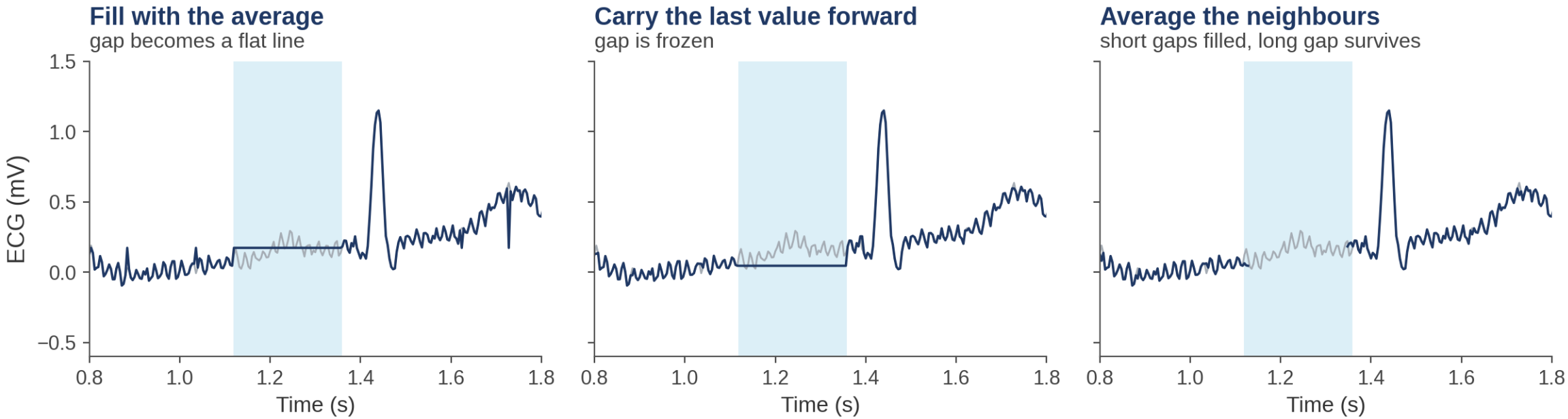
# Option 1: drop the missing samples




Simple, but the recording gets shorter and every beat after the gap moves earlier, so heart rate comes out wrong.

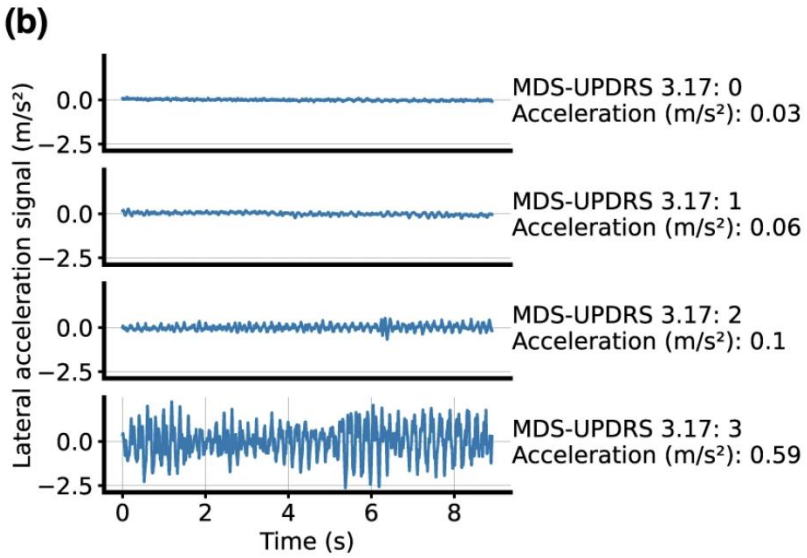
# Options 2 to 4: fill the gap

Filling keeps the timing intact, but each method assumes something different about the missing values.



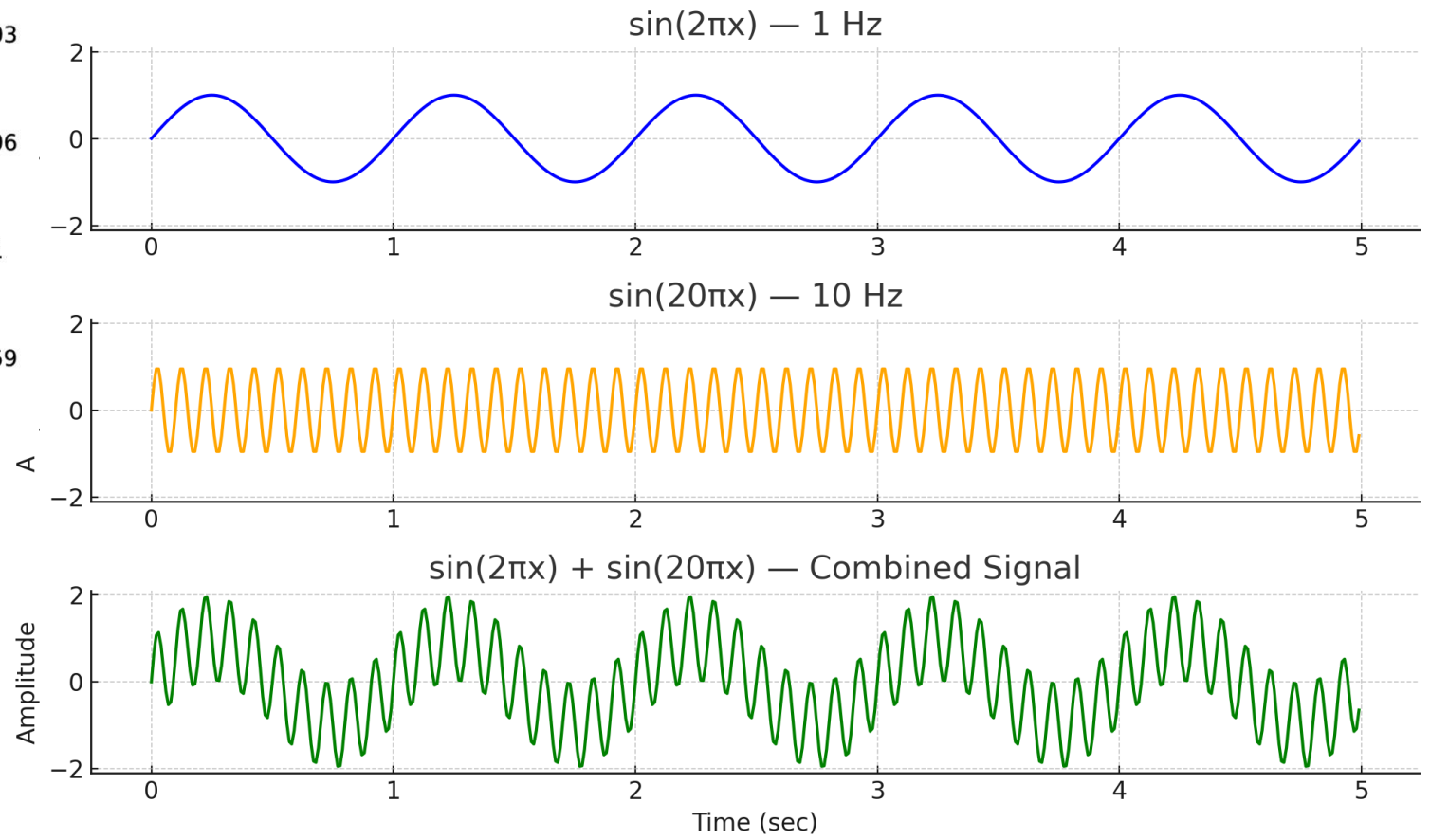

Grey: the true signal. Navy: the result of each method. Blue band: a 0.25 s dropout.

# Revealing Hidden Rhythms in Wearable Sensor Data: Frequency Domain

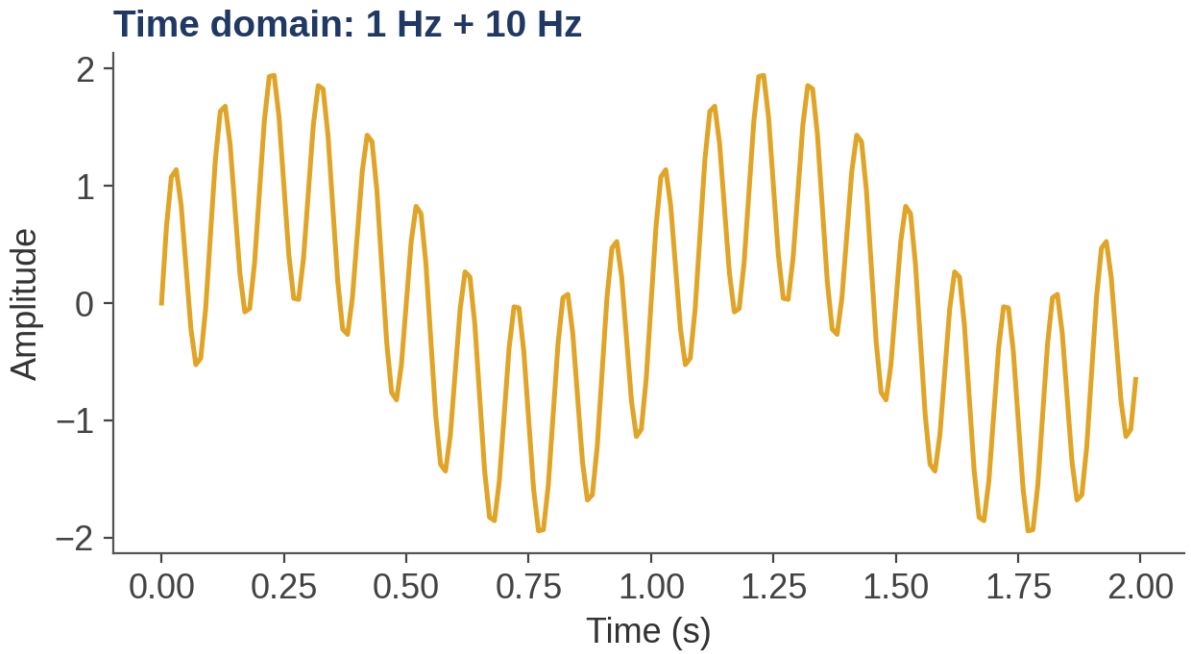


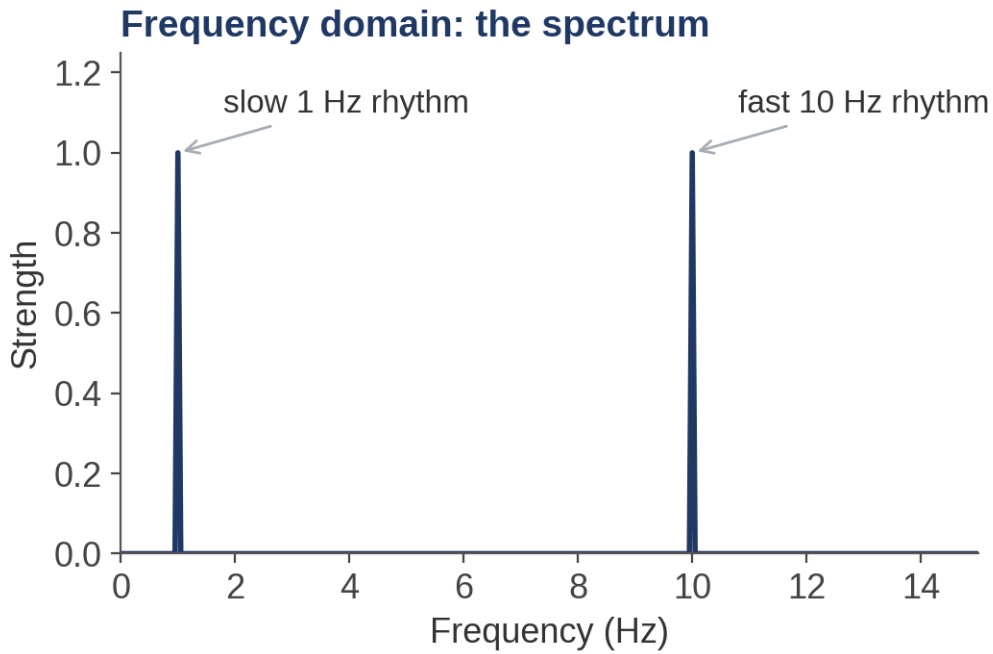






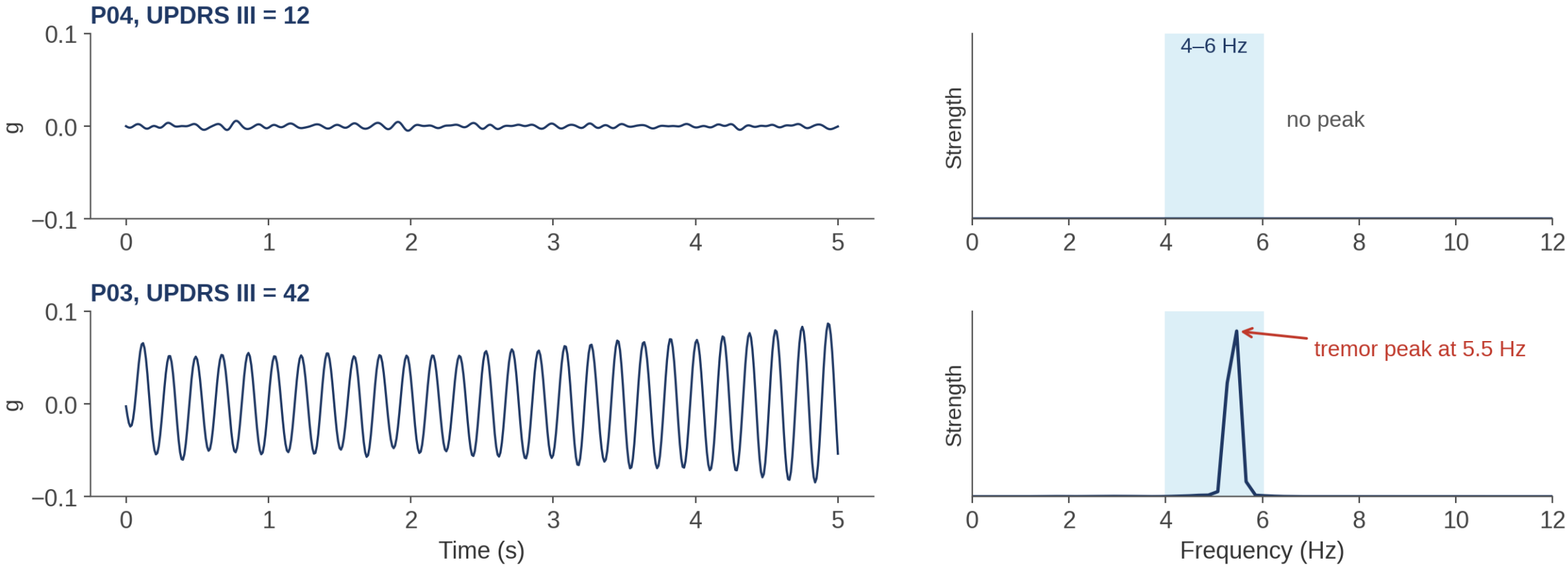
# From time to frequency: the spectrum



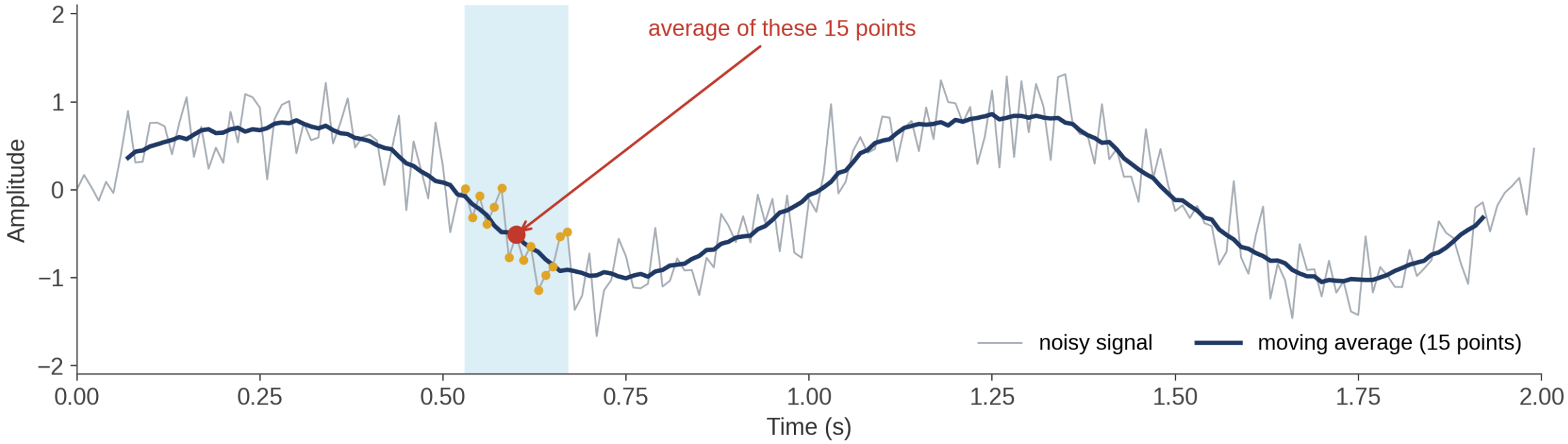

The spectrum shows which rhythms a signal contains and how strong each one is.

# Tremor has its own rhythm



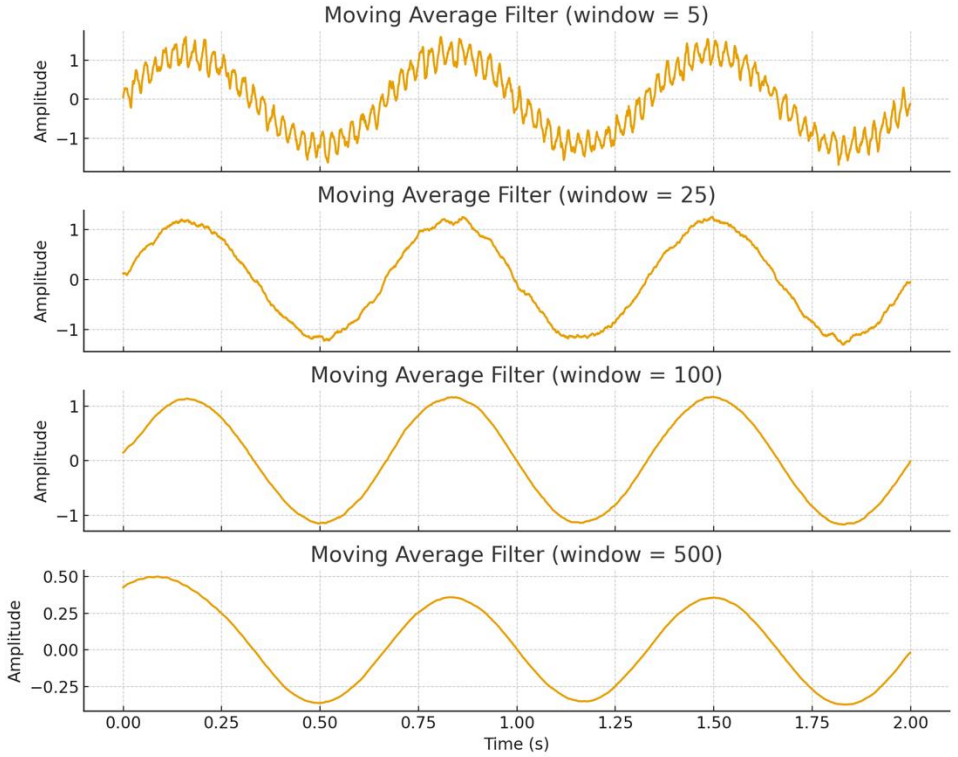

Parkinsonian rest tremor shows up as a clear peak between 4 and 6 Hz.

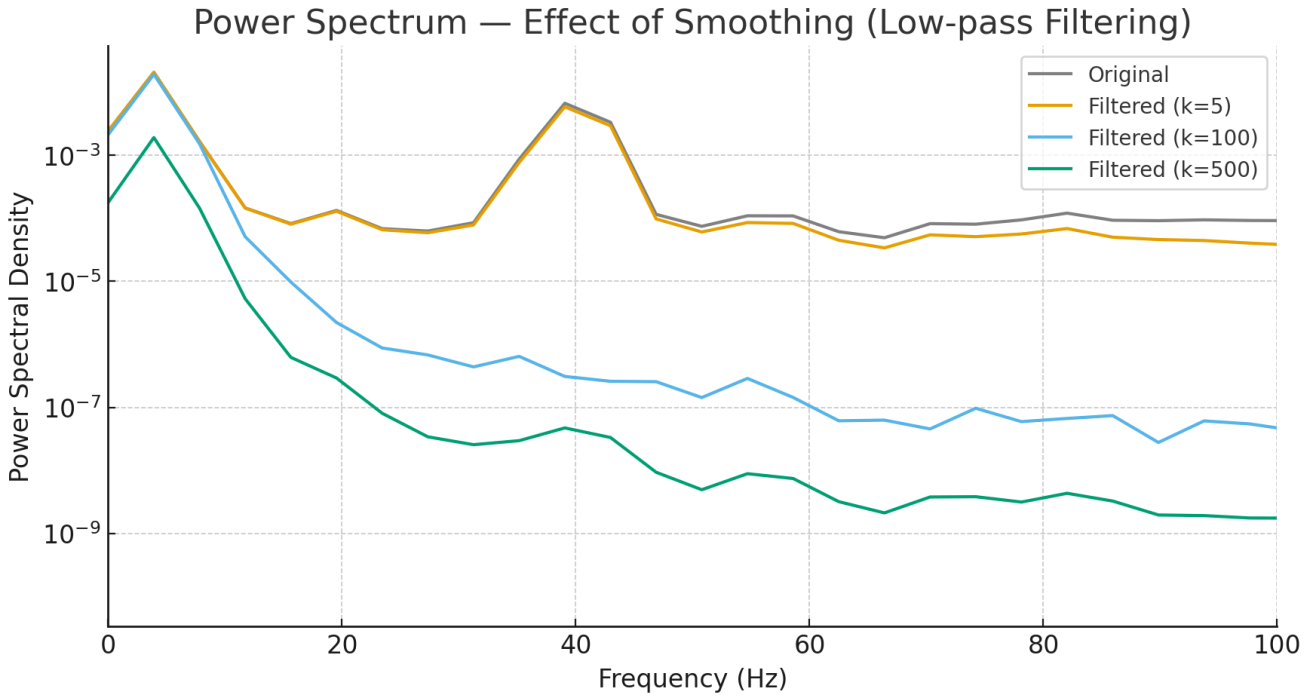
# Smoothing: the moving average



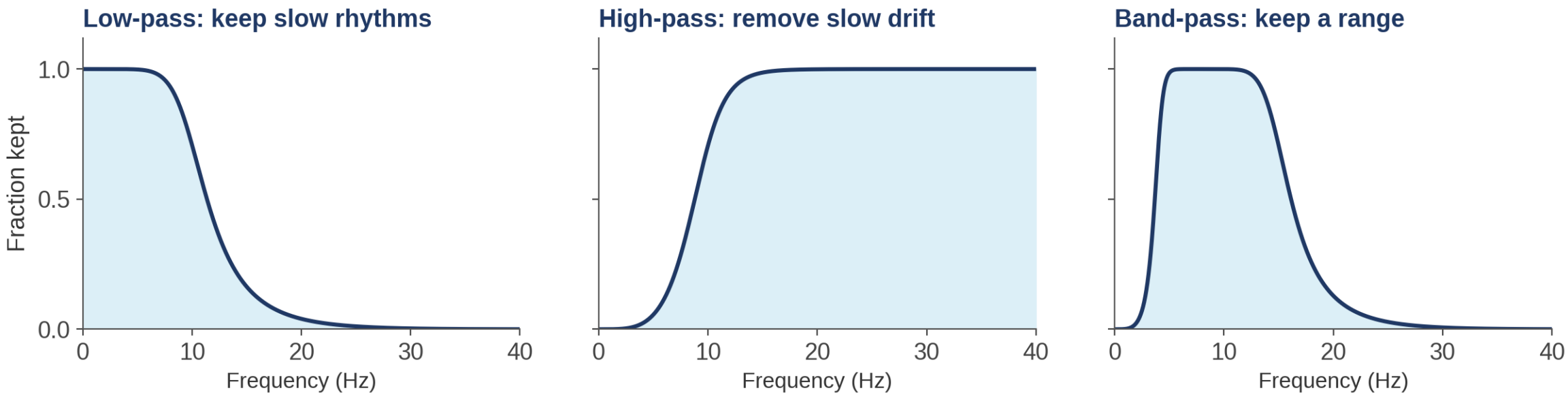

Each point is replaced by the average of its neighbours: random noise cancels out, the slow trend stays.

# Low Pass Filtering: A moving average



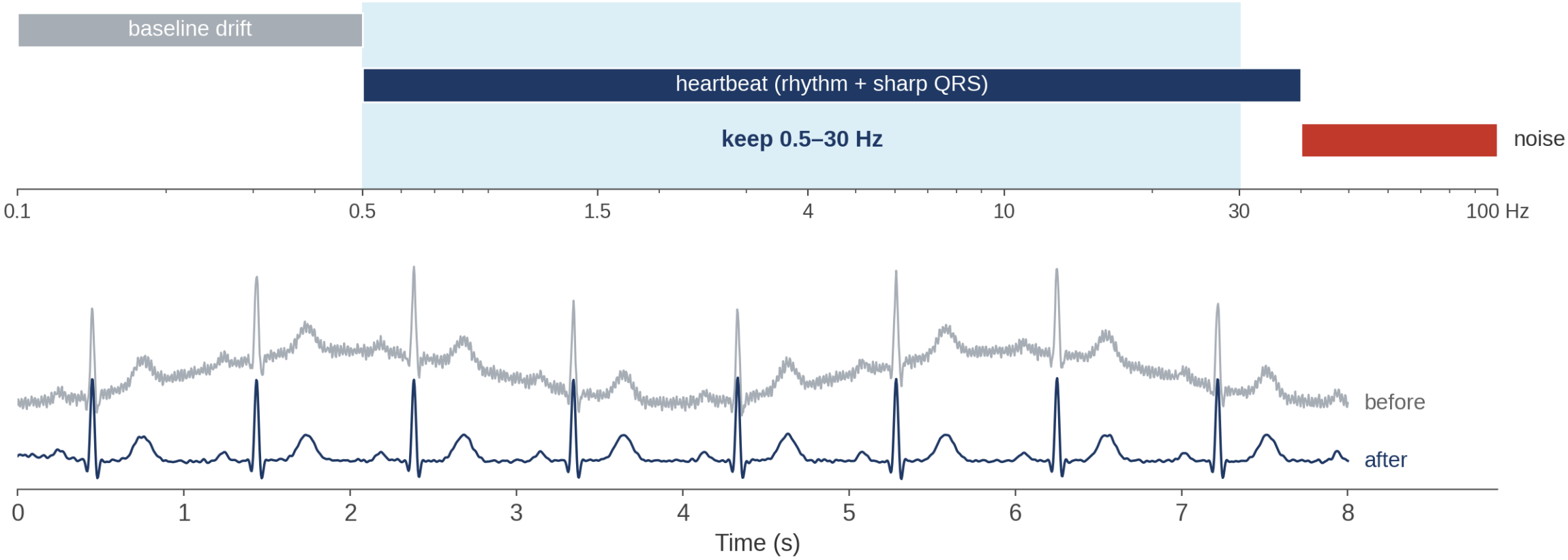



# Filters: choose which frequencies to keep



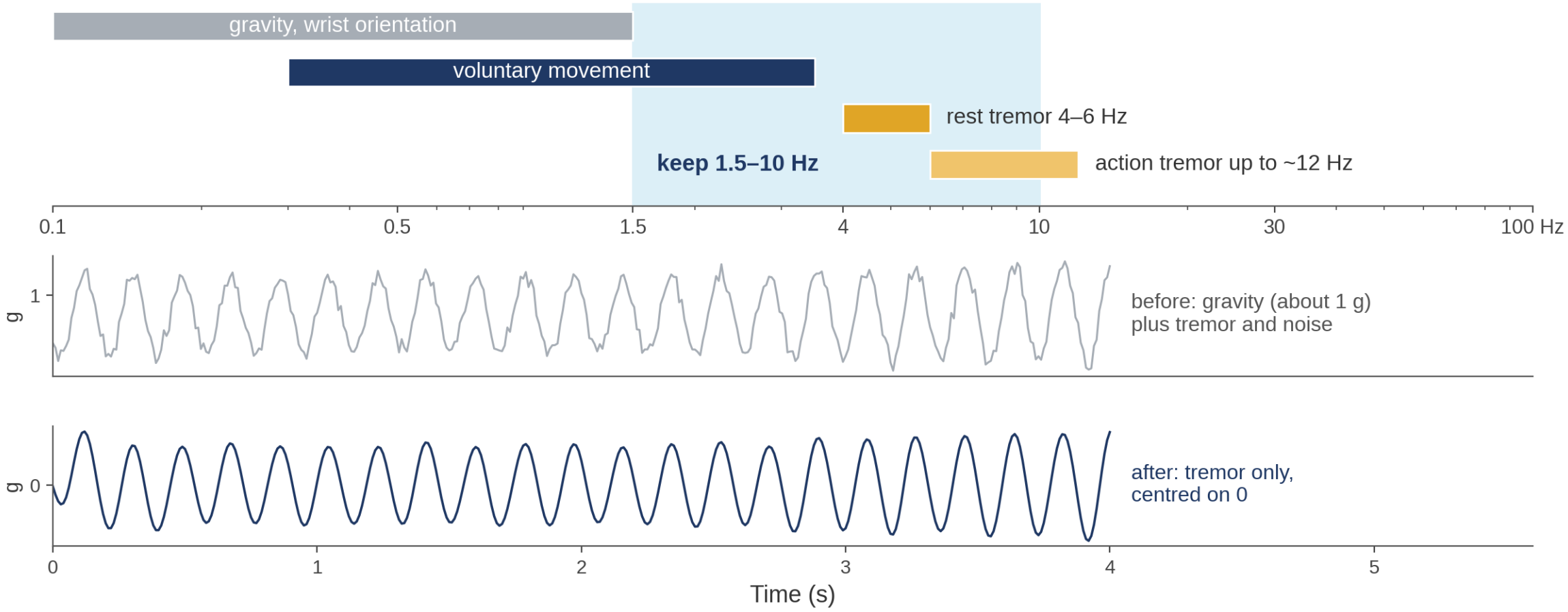

A moving average is a low-pass filter. Removing slow drift needs a high-pass; keeping one range of rhythms needs a band-pass.

# Band-pass for the ECG: remove drift and noise



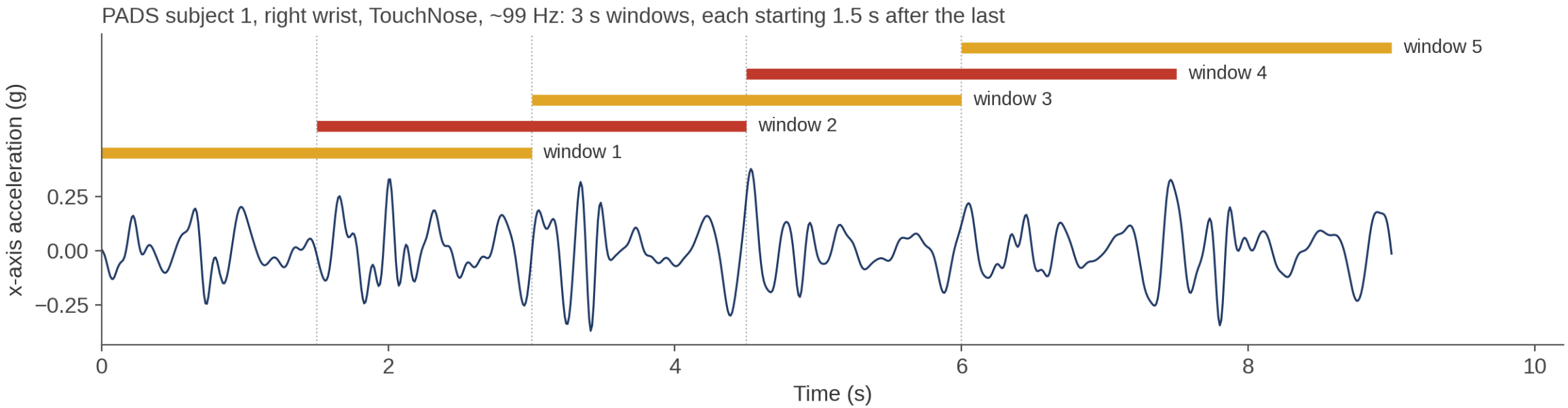

Keep 0.5–30 Hz: the heartbeat stays; the wandering baseline and the noise go.

# Band-pass for the wrist: isolate the tremor



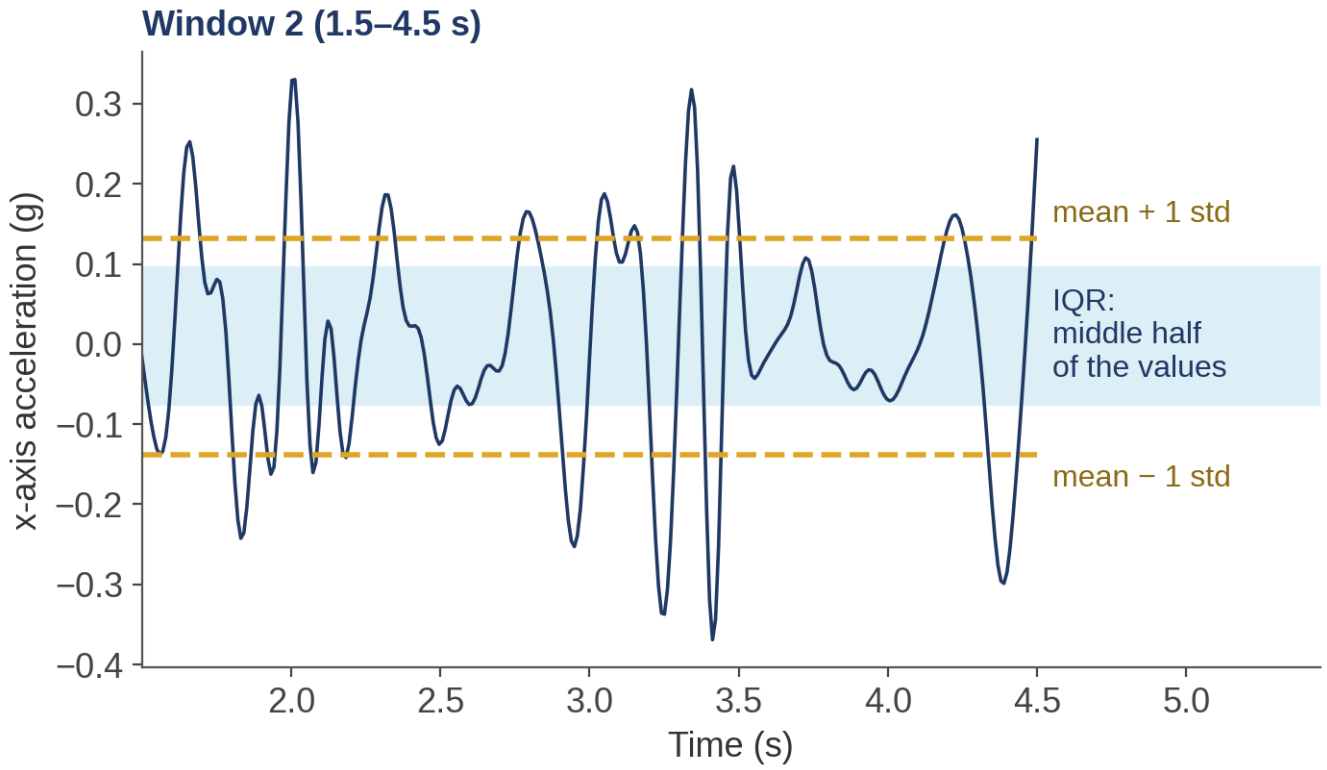

Keep 1.5–10 Hz: gravity and slow arm movements go; the 4–6 Hz tremor stays.

# Windows: cut the signal into short pieces




Signals change over time, so we cut them into short, overlapping windows and analyse each one separately.

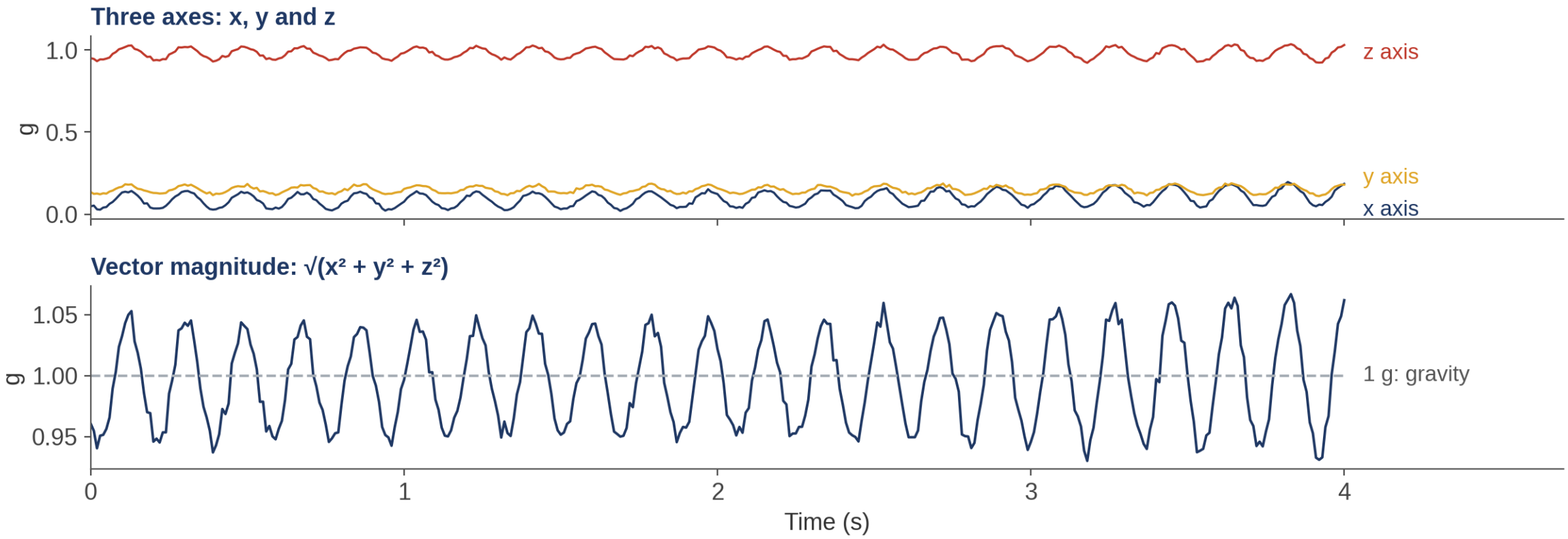
# Features: a few numbers per window




| Window | std (g) | IQR (g) | RMS (g) |
|---|---|---|---|
| 1 | 0.111 | 0.141 | 0.111 |
| 2 | 0.135 | 0.170 | 0.135 |
| 3 | 0.129 | 0.140 | 0.129 |
| 4 | 0.118 | 0.149 | 0.118 |
| 5 | 0.121 | 0.159 | 0.121 |

Spread (std, IQR) and intensity (RMS) turn each window into one row of numbers: the feature table.

# Combining the three axes: vector magnitude




One movement signal, whatever the watch's orientation. A raw sensor at rest reads about 1 g: gravity.

# Let’s Code



**Tutorial 2: work through the notebooks in this order**

1. pandas_intro_EHR.ipynb: tables with pandas
2. TimeSeriesAnalysis.ipynb: ECG and accelerometer signals
3. end_of_day_task.ipynb: link wearable features to health records

https://github.com/Sandoretal/Module_3/tree/main/tutorial_2